Aptiv 2023 Investor Meeting

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OPENING REMARKS

Jessica Kourakos | Vice President, Investor Relations & ESG

Welcome everyone to Aptiv's 2023 Investor Day Live and In Person. It's great to see we have a full house here today in Boston with many familiar faces as well as many new ones. We also know that many others are joining via our video webcast. So, welcome to you as well.

All of today's presentations can be found on our website at ir.aptiv.com under Events and Presentations. And for those of you in the room, we have gone paperless this year. So, a QR code as well as a URL link on our table will direct you to our presentations as well. So, welcome. We have assembled a great event for you today.

Now, before we get started this morning, it is my job to remind you that today's presentations do contain forward looking statements. They're based on our current view of future financial performance and may be materially different from our actual performance for reasons that we cite in our Form 10-K and other SEC filings. So, we ask that you interpret them with that in mind.

Today's presentation has been carefully crafted to address a number of key areas of interest. Taking a look at today's agenda, we start off with our Chairman and CEO, Kevin Clark, who will share our view of the industry and where it's headed and how we plan to usher in the next phase of our business strategy that will drive strong growth and profitability for years to come. Kevin will then be joined by a number of our technology and business leaders, who will highlight our strategy, our portfolio, as well as share insights into our execution playbook and go-to-market strategy in more detail.

Our business segment overview will end a little after 11:00 AM. We will then have about a 45-minute break to allow those of you in the room to have some lunch, view some of the technical demonstrations in the back of the room. And we will provide a countdown clock on the IR website for when we resume at approximately 12:00 PM Eastern. We will then come back with our focus on sustainability.

As Vice President of both Investor Relations and ESG, I'm very proud of the work we do to promote ESG here at Aptiv. Our mission advancing safe, green and connected technologies is so perfectly tied to ESG. And with our event today, we thought that gifting something and donating $10,000 to FIRST would be a great step in that direction. FIRST is an organization that inspires young people to be science and technology leaders and innovators. We've added a QR code in our presentation if you're interested to learn more about this terrific organization. So, on behalf of Aptiv, thank you all for your support and your participation in this event.

Turning back to the agenda. Following our segment on sustainability, we have Joe Massaro, CFO and Senior Vice President of Business Operations, who will provide path to our 2025 targets, as well as provide our long-term framework for sustainable, profitable growth. Kevin will then close with a summary remarks followed by Q&A. For those of you listening remotely, you can submit questions through our event microsite. So with that, let us begin.

[Investor Conference Introduction Video]

STRATEGIC OVERVIEW

Kevin Clark | Chairman & Chief Executive Officer

Thanks, Jessica. Good morning, everybody. Welcome to Aptiv's 2023 Investor Conference. It's great to have you here. It's hard to believe that it's been over three years since our last investor conference back in 2019, actually here in Boston. So, listen, the Aptiv team is very excited to provide you with a detailed update on the industry as well as on our strategy, including an update on our portfolio of products as well as our operating capabilities and how we're confident that, that will translate into long term value creation for our shareholders.

Now, we have a lot of material that we're going to cover today, and I know some of you have already looked through it. There are five themes that I really want to make sure that you focus on and you don't miss.

First, our industry is going to continue to be shaped by the Safe, Green and Connected megatrends that we referenced over a decade ago, are even more relevant today, and are creating massive market opportunities for Aptiv. Second, we have the right strategy to capitalize on these megatrends. To put it simply, we continue to focus on optimizing our business foundation, which enables us to execute on our current business platforms and then positions us to invest in our future platforms. Third, executing our strategy has and will continue to increase our competitive moat and position Aptiv as the only full system solution provider for the electrified, software-defined vehicle. Fourth, we're leveraging these full system capabilities to provide our customers with solutions that deliver better performance at a lower cost for our customers, which distinguishes us from virtually everyone else in the industry. And it uniquely positions us to capitalize on the increasing number of opportunities in both the automotive market as well as adjacent markets. And lastly and most importantly, all of this strengthens our track record of outperformance as a fast growing, more profitable business that's perfectly positioned to deliver outsized shareholder returns.

Now the majority of today's discussions will focus on where we're headed. But I thought it'd be a good idea to begin with a brief recap of where we've come from and how that perfectly positions us for actually what's coming up next. Since our IPO in 2011, we've been leading the narrative about how the automotive industry was going through a significant transformation, really being driven by the Safe, Green and Connected megatrends. And during the last five years, these megatrends have actually accelerated, outpacing our prior expectations, really as a result of increased government regulations and strong consumer demand for more feature-rich, highly electrified vehicles. Over the same period, we've begun to see the adoption of Smart Vehicle Architecture™ with a number of leading OEMs now aligning their next generation architecture approaches to our SVA™ approach.

Now early on, we recognized the opportunities that these megatrends would create. And our strategy was very intentional around designing to capitalize on those trends. We invested in future platforms that would position us for long-term growth and margin expansion by developing very differentiated, high value technology. We enhanced our leadership positions in fast growing current platforms by delivering solutions that optimize both performance as well as cost for our customers. And then, we further strengthened our business foundation, ensuring that we have the talent and the operational footprint needed to execute while continuing our relentless focus on driving cost out of our business, resulting in a much more robust and sustainable business model.

Now, at the very core of our successful strategy is our rigorous process to ensure we see around the corner and anticipate the future challenges our customers will face and ensure we have the solutions to address these challenges. Based on our early experience in electrification and virtually all levels of autonomy, we recognize the complex legacy architectures developed to support ICE vehicles had really reached their limits. So, a fundamentally new approach was needed. In 2017, we introduced Smart Vehicle Architecture™, a flexible, scalable platform that really abstracted software from hardware, something that we're going to talk a lot about today, reducing the complexity and total cost of ownership while actually unlocking new updatable software-defined functionality.

Now, it took a little time for our SVA™ solution to gain commercial traction, but we quickly were awarded advanced development programs with several OEMs in our experience during the evolution of both the Brain and the Nervous System of the vehicle towards a full SVA™ solution has resulted today in over $5 billion of business awards.

Now to capitalize on this transition to SVA™, we very intentionally invested across the entire technology stack, both organically and inorganically. We invested in expanding our offerings in sensors, compute, software building blocks and applications, cloud-based services, which created an industry leading portfolio in Advanced Safety and User Experience. We enhanced our advanced ADAS technologies and built the foundation for our automated driving capabilities. With the acquisition of Ottomatika and nuTonomy which were then further strengthened by the formation of Motional, our joint venture with the Hyundai Motor Group.

We expanded our high voltage portfolio initially through organic investment and more recently with our acquisition of Intercable Automotive. We completed a series of highly accretive acquisitions in engineered components that broadened our portfolio and expanded our presence in adjacent markets. And lastly, we acquired Wind River, which provided a differentiated operating system in middleware as well as the industries first cloud-native DevOps platform, capable of enabling the serverized compute we envisioned with the adoption of SVA™.

At the same time, we've been building a more resilient and sustainable business, enhancing our engineering talent in critical areas, especially software development and systems integration and regionalizing our global footprint which enabled close customer collaboration and the ability to scale engineering, manufacturing and sourcing to lower operating costs, while reducing macro and operational risks.

Our global tech centers are balanced between strategic locations that are very close to our customers as well as locations in best cost countries, including Poland, Mexico and India, where we have access to great pools of talent at very attractive costs. Our manufacturing footprint has enabled us to rotate to an increasingly regionalized and integrated supply chain supporting in-region, for-region production. And we now source the majority of our direct materials in the region where they're actually produced and utilize technologies that allow us to map and model our supply base, helping to mitigate any supply chain disruptions.

The net result of the investments we've made is a business that is very uniquely positioned as the only full system solution provider, with industry leading advanced technologies and operating capabilities that span not only every horizontal layer of the technology stack, but also all of the vertical layers. Over the past five years, we've launched over 60 customer programs with more than 20 customers, including a scalable, hands-free ADAS solution up to Level 2+ for a large global OEM, establishing Aptiv as a first technology provider to successfully deploy a scalable L2+ platform that is now in production. The first infotainment solution with native Google Automotive Services for Volvo and Polestar and best-in-class driver monitoring solutions for multiple OEM systems, including GM Super Cruise. We've also launched over 200 high voltage programs on industry leading global EV platforms from customers such as Tesla, Volkswagen and Volvo. Competitors are focused on a single layer of the tech stack or a single domain. Aptiv is uniquely positioned to operate across the full stack. Our full systems insights allow us to understand the interoperability between and synergies across technologies and domains which enable us to bring them together to create innovative, optimized solutions for our customers.

Our focus on executing our strategy over the past five years has delivered significant value for our customers, which we've translated into outsized value for our shareholders. Our expanding, competitive moat and industry leading execution capabilities have resulted in strong growth in new business bookings across Active Safety, high voltage electrification and now SVA™, capped by another record year in 2022. In our portfolio of Advanced Technologies launch on industry leading vehicle platforms has delivered sustained industry leading growth over market, continued underlying margin expansion, excluding the recent challenges related to COVID and supply chain disruptions and shareholder returns that have significantly outpaced our industry peer group as well as the S&P 500.

While we're proud of the resilient business we've built. We're even more excited about how we're positioned for the future and are confident in our ability to deliver increased shareholder value. I'd now like to move to the road ahead to explain why.

So, just as it did in the past, it starts with the Safe, Green and Connected megatrends that continue to shape the industry and create very large, fast growing addressable markets that are perfectly aligned with Aptiv's portfolio. Growth in the Active Safety market is expected to continue to outpace vehicle production and reach $35 billion in 2030, driven by continued strong demand for highly automated Level 2, Level 2+ and Level 3 solutions. Our addressable market for high voltage electrification is forecasted to increase at a 20% compounded growth rate due to more stringent government regulations and increasing consumer demand.

We expect our addressable market for software applications and building blocks to more than double by 2030 as the industry shifts to software-defined vehicles. And lastly, we now expect nearly one of every four vehicles produced in 2030 to have an SVA-like architecture, creating an addressable market of more than $25 billion for advanced compute in vehicle operating systems and middleware and corresponding Dev-Ops platforms.

We remain confident we have the right strategic framework to continue to drive significant outperformance as we further strengthen the foundation of our underlying business, leverage across Aptiv capabilities to increase our scale and our relevance to deliver high growth, high margin product solutions, and invest in future platforms to see the next wave of new business models and profitable growth.

Now, the next decade will not mirror the last one. Increasing development costs and program complexity as well as the challenges and supporting lifecycle management are impacting today's approach. Our customers' needs are changing, and so must the technology solutions, capabilities and sources of competitive differentiation that shape our strategy.

Beginning with our business foundation, our people remain at the very forefront of our strategy in a rapidly evolving industry, demanding new skill sets, our ability to attract, to develop and engage the right talent is critical to Aptiv's continued success. As we've learned from navigating trade wars, a pandemic and semiconductor shortages, our ongoing success will require flawless execution and supply chain resiliency. And lastly, we've achieved a healthy balance in our regional customer and platform revenue mix. But as customer needs evolve, we have to actually adapt our approach and drive commercial excellence to capitalize on our unique competitive position.

Moving to current platforms. As our recent customer awards reflect, the shift to SVA™ is happening and the acceleration of the demand for battery electric vehicles is providing a further catalyst for OEMs to take a clean sheet approach to vehicle architecture. And as we anticipated when we first launched SVA™, up-integration is occurring across multiple domains and OEMs will be bringing diverse applications together to enable new, more intelligent systems. As a result, we're now developing modular, flexible software applications and building blocks that are scalable across domains, unlike the closed solutions offered by many of the folks in our industry. And we can leverage our entire hardware and software portfolio in the multiple adjacent markets today.

Lastly, moving to future platforms, we need to fundamentally change our approach to enable the software-defined vehicle, moving away from the monolithic software architectures that have historically defined our industry to a cloud-native approach that has become the standard for modern software development today. We're focused on bringing the speed and the agility of cloud-native software to mission critical applications at the edge, where safety and reliability remain absolutely paramount. Automated driving remains a strategic focus for Aptiv, but we also see the opportunity to scale our autonomous technology across multiple applications and use cases, inside automotive and outside automotive. Finally, connectivity continues to proliferate, not just for vehicles, but for all devices that operate at the edge. With vehicles increasingly integrated into a much broader IoT ecosystem. As these devices become smarter and generate more data, more processing and intelligence is required at the edge, creating a need for edge-to-cloud platforms to manage these devices, enable a digital feedback loop and unlock new business models.

Let me now turn to how we're executing this strategy to drive sustainable results. Our strategy is anchored on the next evolution required in vehicle architectures. As I mentioned, in 2030 the car will be part of a broader ecosystem, requiring a cloud-native approach to developing, deploying and operating software throughout its lifecycle and new architecture solutions are really required to unlock this. The need for greater processing power on the vehicle, while at the same time reducing weight in mass is resulting in the up-integration of compute, driving the need to fuse mixed-critical domains, which over time can be augmented with infrastructure at the edge. In parallel, high voltage architecture must also be optimized with increased device up-integration, driving savings in mass, in weight and in cost. But the next major paradigm shift is really coming from software. To truly serverize up-integrated compute, the underlying software architecture must be modernized while still addressing the demands of safety critical real-time applications like we have in automotive and taking full advantage of this software architecture will require the industry to adapt a true DevOps approach. This results in greater speed to market. It unlocks new innovations and enables full lifecycle management in addition to reducing software development, deployment and warranty costs. The next innovations in software architecture will unlock new business models in value creation opportunities for Aptiv as well as for our customers.

Now, to ensure our business foundation is well prepared to support this evolution, we continue to enhance our operating model, enabling us to better meet the needs of both our customers as well as our employees. This means efficiently adding scale and critical skill sets, including software development, high performance compute, AI and machine learning. Now, many of these roles can be located in best cost countries, such as our recently expanded India Tech Center in Bangalore, which is set to grow to 3,000 software engineers by 2025 and will provide a world class experience for our employees there. The product centric roadmaps we're developing are based on a very deep understanding of our customers' needs. The results of extensive cross-functional engagement well in advance of any program quotation. Underpinning this is a robust account-based management approach, empowering our regional businesses with the right local leadership to serve as a focal point for delivering integrated programs, products as well as services. This organizational structure and operating approach accelerates decision making and nurtures very deep relationships at all levels of our customers' organization.

Now, to ensure collaboration and sharing of insights across Aptiv as well as drive a more data driven approach to our commercial activities, we're strengthening the marketing tools and collaboration platforms that we utilize. For example, we successfully rolled out the Net Promoter System across our entire customer base, resulting in tangible, actionable insights across our organization to strengthen our operating model and deepen our customer intimacy. All of this enables the right people with the right processes and the right tools to better serve our global customers.

We have a long track record of ensuring the on-time delivery and quality our customers expect, supported by our global footprint, lean management tools and deep understanding of how to efficiently launch over 2,000 programs each year.

Now, the macro challenges have caused supply chain disruptions over the last couple of years, and that's made that task harder. Disruption increasingly seems to be the new normal. But we successfully navigated this turbulent environment with enhanced tools and capabilities that have improved our ability to anticipate potential disruptions, allowing us to proactively mitigate them and quickly respond when they occur. In doing so, we've made operational excellence and resiliency an even greater competitive advantage, something customers recognize and truly value, especially in the current environment, contributing to our recent record bookings.

Moving to current platforms and starting with the software layers of the technology stack. As I mentioned earlier, we've productized our advanced software offerings, bringing flexibility, modularity and scale to both consumer-facing applications and features as well as the building blocks that enable them. This includes our Gen 6 ADAS platform, a radar-centric, vision agnostic solution which leverages AI and machine learning to increase the availability, robustness and efficiency of the system, resulting in increased flexibility as well as better performance at a significantly lower cost than vision-based alternatives. Aptiv's integrated cockpit solutions that include in-cabin sensing, gesture recognition and advanced audio, seamlessly integrated alongside ecosystem offerings such as Android Auto and our new BMS Software, which will enable faster charging with less battery degradation, improving range and significantly lowering costs for our customers. And all of our software solutions are architected to leverage the cloud for analytics, model training, simulation and testing to drive continuous improvement in a real-time digital feedback loop.

Continuing with the hardware layers of our technology stack, our unique Brain and Nervous System capabilities enable us to develop innovative solutions that reduce complexity and lower total cost of ownership over the life of the platform. In battery electric vehicles, our optimized high voltage solutions reduce weight and mass to improve range while supporting the higher power levels required to enable faster charging. The two most important factors for BEV adoption. We have a comprehensive portfolio of advanced solutions which connect the battery to the grid, including modular busbars, inlets and chargers. And we've expanded that portfolio to include our integrated power electronics, which combines six devices into one. It can host the BMS software that I mentioned previously. These electrified vehicles often come equipped with higher Active Safety and user experience content, which require more advanced and efficient compute solutions to enable. Our centralized computing platforms enable up-integration of distributed single function ECUs across domains. This simplifies manufacturing and supports a flexible, multi-sourcing approach to semiconductors, which improves supply chain resiliency. These optimized platforms are ideally suited for supporting the feature-rich, highly electrified vehicles consumers are now demanding.

Now, as I mentioned, it's important to note that the application of our portfolio of advanced technologies is not limited just to the automotive industry. A number of other attractive markets are shaped by the exact same technology trends and customer needs, including increasing data, power and connectivity requirements, often for ruggedized mission critical applications in a wide range of harsh environments.

In addition to the Commercial Vehicle market, which has obvious technology synergies, our proven software and hardware solutions are supporting end markets such as Aerospace and Defense, Telecommunications and Industrials. Adjacent market revenues now represent 16% of our total revenues and are forecast to have strong growth over the next several years. We also continue to make significant advancements in the development and commercialization of our automated driving technology. We've always viewed automated driving as part of a continuum with Active Safety, which informed our strategy and enabled Aptiv to commercialize our investments across all levels of autonomy. Through our Motional joint venture, we sell our next generation solutions such as imaging radar in Motional's fully autonomous architecture, and use those insights to inform our advanced development roadmap. At the same time, we leverage Motional's technology in our next generation Active Safety solutions. The result is a scalable architecture that spans all levels of autonomy and that we're actively commercializing today. We've launched 10 Level 2 and above ADAS solutions with 8 OEM customers today. And we're actively developing Level 3 solutions for launch in 2025 on our Gen 6 ADAS platform.

In addition, Motional recently deployed the first robotaxi service on the Uber network as part of a 10-year agreement, targeting multiple cities, and will be transitioning to a full driverless service on both the Lyft and Uber networks later this year.

And lastly, our scalable portfolio for autonomy can be leveraged into other markets and applications, including Commercial Vehicle and last mile delivery, two markets where we have active customer engagements today.

Now, as we shared previously, Wind River is a global leader in delivering cloud-native software solutions for mission critical applications across a very wide range of end markets, including Aerospace and Defense, which shares many of the highly regulated, safety critical requirements we deal with in automotive and has already experienced compute up-integration and is now undergoing a transformation in how software is actually developed and delivered. In Telecommunications, which is transforming from proprietary bare metal solutions to cloud-native, software-defined infrastructure to support the roll out of 5G.

We see a significant opportunity to bring the products and solutions Wind River is already deploying in these markets into the automotive market, resulting in the industry's first end-to-end cloud-native DevOps platform that increases the speed and lowers the cost of software development through a highly automated build and test environment, streamlines deployment of that software to vehicles with unique edge software products, including the only real-time operating system that supports containers and microservices across all levels of safety certification and enables full software lifecycle management through full time analytics, monitoring and orchestration capabilities. We're deeply engaged with several customers regarding Wind River solutions, which we're confident will lead to commercial awards over the course of 2023.

Over the long term, this unique edge-to-cloud portfolio positions Aptiv to capitalize not only on the automotive industries transition to software-defined vehicles, but also on the digital transformation of multiple industries as intelligence increasingly moves to the edge. We'll be able to leverage our full stack capabilities all the way from advanced sensors, like radar, to highly automated software-defined vehicles, aircraft or machines, to the virtualized 5G infrastructure, managing the data to and from these assets to the cloud where our DevOps platform will be used to analyze this data, make performance improvements and deploy and orchestrate the next release of software. As a vehicle and other devices form an increasingly connected IoT ecosystem, our edge-to-cloud portfolio opens up new business model opportunities and positions us for sustained, long-term profitable growth.

So when you step back, we see that the needs of our industry are rapidly changing and the optimized, cost-effective solutions that address those needs will increasingly require full stack capabilities that extend across all domains and up the stack. As a result, Aptiv's advanced technologies and operating capabilities remain perfectly positioned to solve our customers' toughest challenges from architecture and hardware to software and the cloud.

And our full systems portfolio will continue to deliver innovative solutions that no other industry player can provide, further increasing our competitive moat. Offerings that remain focused on individual layers of the tech stack or specific vehicle domains will increasingly be challenged, as architectures move towards of serverized cloud-native, software-defined approach.

Given our Brain and Nervous System capabilities, we have a higher content per vehicle opportunity and a larger addressable market, which provide a significant tailwind for growth. With the continued penetration of electric vehicles, ADAS and SVA™ and limited de-contenting for Aptiv from these trends as well as the portfolio enhancements we've made in high voltage electrification as well as software, we now expect our total addressable content per vehicle to increase more than 50% to over $2,300 by 2030, resulting in our core addressable market increasing more than 70% to over $200 billion. And as shown on the right by leveraging our technologies into attractive adjacent markets opportunities, we have the ability to nearly double our addressable market opportunity to $400 billion.

So in summary, we have tremendous momentum and the wind is at our back. As a team, we remain laser-focused on executing our strategy, delivering on our commitments, and continuing our track record of outperformance, enabling us to realize our vision for the company.

We're on a path to $40 billion of revenues and 17%+ operating margins by 2030, driven by volume leverage and continued robust growth over market, benefits related to our strong competitive moat and further optimized cost structure and increased mix of higher margin software and solution sales with significant additional upside from continued disciplined capital deployment. All of this is enabled by a very committed management team that has created a winning culture. The output of a company that reinvests in its people, its processes, and its technology portfolio and makes Aptiv a more sustainable business, delivering long-term value creation for all our stakeholders.

So, as you can imagine, it's a very exciting time to be at Aptiv. So, to further explain why, let me introduce you to some of the key members of the management team responsible for executing our strategy day-to-day who are presenting today. First, Matt Cole and Bill Presley will provide an overview of our AS&UX and S&PS segments, followed by Christian Schäfer, who will highlight how our capabilities in the Brain and Nervous System of the vehicle result in an optimized SVA™ hardware platform. Glen De Vos and Avijit Sinha will then cover our future platforms designed to enable the cloud-native vehicle. Josie Archer will then explain how our initiatives related to commercial excellence are closely connected to our technology road map and are enabling Aptiv to realize the value we create.

Following a short break for lunch and an opportunity to see our technology demonstrations, you'll hear from Obed Louissaint and Kate Ramundo on our sustainability initiatives. Joe Massaro will then cover how this all translates into the financial framework, that delivers value to our shareholders. I'll then do a short wrap up before we open it up for questions.

So with that, I'd like to welcome my colleague, Matt Cole, to the stage.

AS&UX: ADVANCING THE BRAIN

Matt Cole | Senior Vice President Advanced Safety & User Experience

Thank you, Kevin, and good morning.

As Kevin mentioned, no one is better positioned to enable the software-defined vehicle of the future than Aptiv. Aptiv is uniquely established as a full system solutions provider with the Brain and Nervous System with strength across all key domains of the vehicle's architecture. AS&UX represents the Brain in this architecture. This encompasses our deep expertise in centralized computing platforms, advanced safety systems, and in-vehicle user experience. The portfolio is enhanced by a number of core capabilities, such as AI and machine learning and the cloud-native software approach. These key capabilities expand our competitive moat and differentiates our solutions. Core capabilities are then translated into productized solutions, well-defined configurable offerings, which significantly increase both software and hardware content. As we expand software content, we're also focused on up-skilling our talent and ensuring the necessary scale and best cost countries to drive improved competitiveness.

Now, Glen and Avijit will provide detail on the power of Wind River later this morning, but it's important to know two things. Wind River further differentiates our products and platforms. And secondly, as important, the adoption of Wind River tools significantly improves our internal software development efficiency. Engineering efficiency is a critical lever for AS&UX operating margins. The combination of our unique offerings, full system level capabilities and commitment to operational excellence is driving sustainable, profitable growth.

Let's dive deeper into growth. Our growth is propelled by the increasing need for high performance compute, adoption of advanced vehicle architectures and the transition to a cloud-native software platforms. The acceleration to battery electric vehicles is requiring OEMs to rethink their architecture, creating a natural catalyst for the adoption of our Smart Vehicle Architecture™ and software-defined vehicle solutions. Our advanced technologies and perception, compute software and tools are all coming together and it couldn't have happened at a better time.

As a result, we see significant opportunities with domain and zonal controllers and software, both middleware and apps. We forecast our total addressable market to more than double by 2030. Execution of our long term strategy has positioned us well and we are leading the market in all key growth areas. AS&UX has the right business foundation to efficiently support the strong growth we see ahead. The strength in our organization starts with our people and a culture of innovation combined with extensive domain expertise across the vehicle.

Operational excellence is driven by our relentless focus on optimizing our cost structure, both engineering and manufacturing, and improving supply chain resiliency. Also, our global engineering organization is driving efficiencies with reusable design blocks and platforms. We are laser focused on having the right capabilities, the right capacity and a competitive footprint. In fact, I just recently returned from the grand opening of our State of the Art Technical Center in Bangalore, India, where we are significantly expanding our product development activities, targeting 50% growth over the next couple of years. Additionally, we are continuously enhancing our supply chain resiliency, establishing more control over our supply base. We have completed more than 100 redesign projects as well as positioned ourselves to be significantly more hardware-agnostic.

As Kevin mentioned, we see a future where up-integration of compute and the abstraction of hardware from software will allow for semiconductor chips to be significantly more interchangeable, which drive improvements in our resiliency as well as resiliency for our customer base. Additionally, we have a regionalized manufacturing footprint to capture the value of sourcing and producing in the region. Lastly, on commercial excellence our record bookings last year speak for themselves and we continue to up-skill our customer facing teams to support the customer with more integrated and higher margin solutions they require.

Moving to our product portfolio, our industry-leading platforms enable the feature rich, highly automated vehicles consumers are demanding. We are pioneers in providing exterior sensing and software products for our Active Safety systems with more than 20 major OEMs around the globe. In User Experience, we are an industry leader for interior sensing and integrated cockpit solutions, leveraging open platforms which support increasing levels of safety, comfort and convenience. And as our business model continues to evolve, we have established a new product category, Smart Vehicle Compute and Software. This includes high performance compute, zonal controllers and centralized compute as SVA™ is happening now. Just look at the $5 billion of new business awards we have to date in this category. This product category also includes Middleware and DevOps through Wind River's cloud-native software platforms, rounding out our full stack capabilities and enabling the software-defined vehicle. We have already had an early success with Wind River, with their first win with Geely, a large Chinese OEM for the next-gen Level 2+ Active Safety system provided by Aptiv. This product line provides high growth, margin accretive offerings and supports new business models. These advanced technology solutions are enabling the future of mobility, and they can absolutely be leveraged in adjacent industries undergoing similar transitions. Our expertise in automotive, hardware and software solutions can apply to other markets, such as commercial vehicles, aerospace and defense, last mile delivery and telecom.

Diving deeper into Active Safety, we continue to expand our competitive lead with ADAS with productized solutions and advanced software offerings. At CES this year, we demonstrated the capabilities and benefits of Aptiv's radar-centric, vision provider agnostic Gen-6 ADAS system. This evolution leverages our latest 4D radar technology for leading perception capabilities and high performance in all types of driving conditions. We also use AI and machine learning to increase the availability, robustness and efficiency of this system. Why is this important? It reduces the need for expensive, high-end vision technology. This is the most cost effective approach for Level 2, Level 2+ and Level 3 systems, as much as 25% lower cost compared to vision centric systems. Let's watch a short video showcasing our Gen-6 ADAS platform solution.

[Gen 6 ADAS Platform Video]

Our investments and scalable approaches to advanced safety solutions are not only seeding the next wave of growth, but they're also helping to drive the democratization of advanced safety systems. Aptiv's flexible architecture approach has been a game changer in the industry and has helped OEMs deploy Active Safety solutions across their multiple vehicle platforms. As Kevin mentioned, we have launched 10 level 2 and above ADAS solutions with eight OEM customers. This includes a recently delivered full stack scalable ADAS platform capable of supporting a scalable Level 2+ platform for a large global OEM. So, what does this mean for Aptiv? It presents significant content growth opportunity with growth through Level 2+. And that has really been our sweet spot. But we anticipate a step change in content as we get to Level 3 and above. Our unique strategy with Active Safety offerings and Motional's fully autonomous solutions is to commercialize our investments across all levels of autonomy.

Turning to User Experience, we have a track record of bringing exciting solutions to our customers. First to market with an integrated cockpit controller, first to market with Android Auto Solutions, and first to market with In-Cabin sensing features such as gesture recognition. We continue to invest in User Experience to allow OEMs to realize the benefits of open platforms over their updates and integration of adjacent domains. As ADAS technologies improve safety, we see the interaction with In-Cabin User Experience as critical to enhancing the adoption and use of these technologies.

As experts in both Active Safety and User Experience, we are focused on developing intelligent solutions which ensure the driver is situationally aware and technology confident regardless of the level of autonomy. These solutions are seeding the next wave in User Experience with growth underneath and tremendous customer pull. The new features and functionalities in vehicles today have created significant ECU complexity and distributed intelligence that is both unaffordable and inflexible. Through the up-integration of hardware, we are significantly reducing complexity, lowering total system cost for our customers, but resulting in more growth for AS&UX. This is increasing Aptiv's share of the wallet as wealth transfers from single function ECUs to larger centralized compute systems in the vehicle.

One example of the power of Aptiv is reflected in a major new business award from last quarter. We secured a high volume award with a customer in Asia to provide our next generation integrated cockpit controller. This solution will be launched across their entire EV portfolio, one solution across more than five vehicle platforms. However, one key reason why Aptiv was selected was due to our ability to provide driver monitoring and interior sensing features after – over there, after the vehicle was sold. Cross domain experience is becoming important for our customers as we shift towards a software-defined vehicle and new architectures.

Our Smart Vehicle Architecture™ solutions not only enable a fully optimized architecture, but a cost effective solution for our customers. This approach has been validated in a study by one major OEM demonstrated that hardware up-integration would reduce ECU complexity by 75%. That's 75% less ECU complexity resulting in higher performance, higher quality with lower total system costs. What that means is that Smart Vehicle Architecture™ is not only great for our customers, it leads to a larger, faster growing and more valuable business for Aptiv. We've demonstrated our leading solutions and high performance compute, combined with capabilities across all domains, provides us with a competitive advantage.

Scalable platforms requires scalable software, and that's what's driving us to containerization, virtualization in a cloud-native approach. Only Aptiv and Wind River bring this approach, end-to-end DevOps capabilities across all key domains. Let me give you an example. When moving to higher levels of automation, specifically Level three, the simple task of driver handoff touches at least six legacy domains, which all must interact to provide a safe and comfortable experience for the driver. This can only be done efficiently through containerization and virtualization, core capabilities of Aptiv and Wind River.

The ability to bring together and manage full stack solutions is critical for OEMs. Our end-to-end software development and deployment capabilities enable reduced time to market, deployment of new features and seamless over-the-air updates. When you look at our portfolio, we understand everything on the vehicle from sensors all the way to the cloud. This is something that no other supplier can offer. Multi-domain Tier 1s understand their applications, but have limited middleware offerings and often must partner with others. Proof Aptiv wins, we were selected for our ability to add ADAS and comfort features to an integrated cockpit controller after the sale of the vehicle by leveraging our cross domain experience.

SOC providers typically have single domain-specific hardware and software expertise. However, they're unable to capture Smart Vehicle Architecture™ opportunities. Proof Aptiv wins, we have $5 billion of SVA™ wins with seven OEMs to date.

Software and cloud providers often have a narrow focus and serve a wide range of industries. Proof Aptiv and Wind River wins, our early success in leveraging Wind River solutions for our Level 2+ ADAS safety system for Geely. It's clear Aptiv has an inherent advantage in these advanced solutions.

All of this is leading to is driving sustainable, profitable growth. Our strong bookings and top-line growth in highly differentiated technology areas is translating to strong margin expansion, which Joe will walk through in much greater detail. From now to 2025, we are expecting sales growth of 14 points over market. Strong growth will continue in Active Safety and in Smart Vehicle computing software, we will begin to see SVA™ bookings translate into revenue.

As legacy programs in User Experience roll off and our next generation platforms ramp up, we do see accelerated growth beyond 2025. With strong flow through on this revenue growth and by maximizing our operational levers, we have a clear path to over 13% OI by 2025.

We see margin expansion from product mix, especially as Smart Vehicle compute and software grows as a percent of the overall business. Additionally, we are driving significant engineering efficiencies through the expansion of engineering in best cost countries as well as the adoption of Wind River tools. By leaning out our operations and improving our supply chain resiliency, we will see favorable impact to our margin profile. We are executing on our strategy, making mobility safer, greener and more connected. I'm excited about the future of AS&UX and I'm excited about the future of Aptiv.

I will now pass the baton on to Bill Presley to talk about exciting opportunities in our Signal & Power Solutions segment. Bill?

S&PS: OPTIMIZING THE NERVOUS SYSTEM

Bill Presley | Senior Vice President & Chief Operating Officer, President Signal & Power Solutions

Thanks, Matt, and welcome, everyone.

I'm excited to be here with all of you today because I'm very excited about the future of Aptiv, because as economies around the world move to smarter and greener solutions, Signal and Power Solutions positioning and value proposition in the Nervous System of the vehicle has never been greater. Our unique portfolio and capabilities are making Signal and Power Solutions the partner of choice for OEMs as they transition to optimized, electrified architectures and make them available to consumers around the world.

Increasing levels of technology and feature content are creating product and process complexity that only Signal and Power Solutions is capable of addressing at scale. It's very clear that our holistic approach sets us apart. We are the only supplier that can deliver full system solutions that are optimized for size, weight and total system cost from grid through vehicle. And as you'll see here shortly, Signal and Power Solutions has continued to expand the competitive moat. We're bringing to market new product offerings in Integrated Power Electronics, as well as Battery Management Systems.

In addition to high voltage busbars and complementary interconnect solutions, that technology blends perfect with Aptiv's Smart Vehicle Architecture™, which greatly simplifies architectures and allows for a higher degree of automation both in our plants as well as OEM factories. Ultimately, this translates into growth with accretive margins as we leverage our global scale to deploy our products and processes across a broad mix of customer platforms and OEMs.

Signal and Power Solutions is perfectly positioned to benefit from the accelerating market trends in our industry. Our EV penetration assumptions have increased since our electrification teach-in in June of 2021. And we now expect that by 2030, 50% of the vehicles on the road will have high voltage solutions in them. That's double the penetration we see today and conservative relative to IHS.

In addition, consumers want greener, more technology-rich experiences, and that means greater feature density is required in the electric architecture of the vehicle. So a couple of things are going to have to happen. Architectures are going to have to get smaller to physically fit in more functionality. They're going to have to become lighter to enable greater range, and they're going to have to enable more sustainability.

Now, most OEMs were working around legacy portions of the vehicle architecture when they worked to create their first electric vehicle. And that meant compromises had to be made as they work to layer in electrified powertrains into internal combustion engine platforms. Now, while those compromises are understandable, that approach leads to significant architecture inefficiencies. Wire lengths were increased by about 10%, which yields a 30% weight penalty on a vehicle where weight is directly proportional to range. Now the OEMs know that those inefficiencies need to be removed to make electric vehicles profitable.

And the path to profitability is through systems integration, specifically tailored to electric vehicle applications and to reducing battery cost with the assistance of advanced Battery Management Systems. Integration will result in electrical architectures in vehicles that are lighter and lower cost from an overall system standpoint. That means that the next generation of electric vehicles provide the natural break-point for OEMs to rethink their architecture. And Aptiv's Smart Vehicle Architecture™ is the answer for customers that need full system solutions that are optimized for weight, size and total system cost from grid through vehicle.

Now, S&PS is the only supplier with the expertise, full system solutions, and business foundation to address the biggest challenges our customers are facing today at scale. So we thought it would be best to demonstrate the complexity of what we do by giving you a view into the day in the life of one of our manufacturing facilities. So we have a state-of-the-art plant in Serbia that I use as an example that services a European OEM. Every day that plant takes 2,700 raw material part numbers to produce 600,000 cut leads. They take those cut leads and they assemble them into wire harness assemblies containing 1,600 circuits, and they do this on over 200 meters of final assembly stations. So you have an appreciation for scale, if you put all the final assembly stations end-to-end they would be about three football fields long, and that's for one vehicle harness set.

Now the harnesses are ordered by vehicle identification number, which means we get a feature list that has to be distilled into a buildable harness combination. And there are over 1.5 million build combinations. And we have a very short timeframe to execute all this. From the time we get the harness ordered to the time it has to show up on the dock at the customer's factory is five days. And globally, we have to execute with quality and agility every day because our engineers and manufacturing experts process over 40,000 engineering changes per year. And we accomplish all of this while launching over 1,600 programs per year. So with this level of complexity, it's critical that we have the right people in the right place with the right tools.

So we design and engineer in close proximity to our customers technical centers to ensure that we have aligned solutions. And then we manufacture in locations that provide the OEMs with the best possible landed cost at their assembly plant. So simply said, we design locally and then we scale globally to support our customers with optimized full system solutions anywhere in the world.

As demand for electrification rises, customers are increasingly looking for partners that can help them pack more content and functionality into smaller, lighter components that reduce space consumption and weight, while also enhancing the performance and reliability of the system. Signal & Power Solutions generates approximately $13 billion in revenue per year. We are one of the top electrical distribution system suppliers in the world. We're the second largest automotive interconnect provider and we're the top provider of cable management and fastening solutions.

So customers would have to engage with three to four different suppliers and commit time and engineering resources to develop the same full system solutions that comes with engaging with Signal and Power Solutions alone. As an example, a customer came to us and asked us to quote a design that they had specified. We analyzed that design with our engineering tools. We used our product and process library to modify the design, and then we vertically integrated the components through our Signal and Power Solutions component catalog. We went back to the customer with a design that was 10% smaller, 30% lighter at a greater than 10% cost reduction for the overall system than what they had asked us to quote. That's a perfect example of how we provide value to the OEM and increase our content per vehicle while displacing our competition.

So given our expertise in automotive and the harsh environment applications, we also have natural adjacency in other end markets. And we continue to grow in Industrial, Aerospace and Defense, Telecommunications and Commercial Vehicles, just to name a few.

Our portfolio, coupled with our local presence and global scale, has perfectly positioned Signal and Power Solutions to benefit from the shift to electrified vehicles. And we continue to invest both in our high-voltage and low-voltage solutions to transform our business and our products to deliver for our customers. Now let's take a deeper look into some of our newest product offerings right after this video.

[Powering the EVs of Today and Tomorrow Video]

OEMs are looking to save space in their electric vehicles while needing to transfer more power throughout the vehicle and those things are counterintuitive; usually more power means bigger. So that problem requires a very innovative solution, and the answer to that problem comes in the shape of flat form conductors called busbars.

Intercable offers a comprehensive, market-leading portfolio of insulated and overmolded busbars, with the complementary interconnects that address the challenges in the battery pack environment. They assemble these components into very technically complex modules.

As the animation in the video showed, the process and product knowledge required to design and manufacture these types of products and busbars are also integral to Aptiv's Smart Vehicle Architecture™. SVA™ couples the busbars with a modular zonal architecture and our Dock and Lock™ connection systems to greatly simplify the overall system design. Christian Schäfer is going to come up here shortly and give us more detail on that.

These technologies form the foundation of the next generation of electric vehicles. Intercable expands Signal and Power Solutions portfolio of high voltage power distribution solutions while utilizing Signal and Power Solutions global footprint and customer base globally. The combination of Signal and Power Solutions, global scale and customer presence, combined with Intercable's newest generation of technology, will allow us to bring this space and weight savings technology to all of our customers around the world.

So at S&PS, we engaged with customers on the next generation of electric vehicle architectures and we saw a couple of things. We saw that power electronics were widely distributed throughout the vehicle, taking up valuable space and that current Battery Management Systems weren't enabling a reduction in battery size. Our customers recognized Signal and Power Solutions for systems expertise and asked us if we could help address these challenges. So we accepted that challenge. And we brought to market new product offerings in hardware and software that have unique selling points that provide value for our customer while widening the competitive moat at Signal and Power Solutions.

Our Integrated Power Electronics and Battery Management Systems labs are now fully operational. They didn't exist 24 months ago. We have over 150 engineers and scientists in disciplines in mechanical, electrical and chemistry engineering as well as software developers working in those labs. We've invested over $20 million since 2021 on engineering, development and testing capabilities.

And as a result of those organic efforts, as you heard Kevin mentioned earlier, Signal and Power Solutions developed an Integrated Power Electronics module that integrates six unique functions into a single system solution, in the past each of those functions would have been a separate module. Our Integrated Power Electronics solution allows our customer to enjoy a 33% space reduction for the same level of functionality in the vehicle.

In addition to power electronics, we're developing the Battery Management Systems that our customers challenged us with. We're utilizing our software knowledge and capabilities to develop a Battery Management System that will reduce battery degradation while increasing charging speeds. It will also improve battery state estimations, which will ultimately improve battery safety. All of this improves the battery life, which allows for a size reduction, which directly increases the vehicle range and lowers the overall cost of the battery pack for the OEM.

Now customers have been very receptive to our new product offerings. We were very recently awarded our first power electronics assembly with integrated Battery Management System with a major North American OEM. Power Electronics and Battery Management Systems are a natural complement to our well-established, multi-voltage platform solutions and our first revenue with these exciting new products starts in 2026. We estimate that by 2030, these products will generate over $0.5 billion in revenue with accretive margins for Signal and Power Solutions.

So as we saw earlier, electric vehicles provide a natural introduction point for Aptiv Smart Vehicle Architecture™. SVA™ is an optimized solution, which reduces the weight, length and complexity of the wire harness system and it creates physical space in the vehicle to literally add more content. It also directly increases range by shedding the weight penalty. SVA™ is good for Aptiv's customers. It reduces the component complexity and gives them a more efficient system in terms of total cost, weight and size with the opportunity for them to reduce manual labor in their own assembly plants. SVA™ is very good for Aptiv, because Aptiv products enable the optimization of the architecture within the vehicle, which provides us with more addressable content.

During our electrification teach-in in 2021, our ICE to BEV addressable content increase was more than double with ICE having approximately $500 worth of content per vehicle and BEV having approximately $1,200 of content per vehicle. Now with the additional content opportunities unlocked by the portfolio expansion I've just told you about and the increased demand for higher content vehicles, we estimate an addressable market of approximately $800 for an ICE vehicle, while our BEV content per vehicle is almost triple at $2,300 per vehicle. Now, this is a very important point. Even though vehicle content for Aptiv is going up, the overall cost goes down as systems are optimized through integration and utilization of full system solutions.

So let's just step back for a moment and look at the competitive landscape, because this is where a picture frames it better than any words, okay. We have purposely and methodically developed our Signal and Power Solutions portfolio to be a full system solutions provider from grid through vehicle. And as we expand into areas such as Integrated Power Electronics and Battery Management Systems, Aptiv has the unique ability to leverage the broader compute and software capabilities to tailor the system performance in ways that literally no one else can. Signal and Power Solutions will remain relevant on a large scale, as rationalization and integration happens in the industry, which means Signal and Power Solutions is positioned for sustainable, profitable growth.

Trends in electrification are accelerating, and our portfolio and execution capabilities are unmatched in the industry. The momentum created with our portfolio and expertise is supported by a relentless focus on quality, as well as reducing structural costs while driving manufacturing performance. Over the next three years, we will drive margin improvement through material, manufacturing and engineering productivity. Signal and Power Solutions will see an increased mix of higher margin products like the high voltage solutions we saw here today. We're using our state-of-the-art engineering tools to maximize data reuse and reduce development time.

Higher manufacturing yields are being realized through the utilization of digital systems to increase our overall equipment effectiveness. By collecting real-time data on the plant floor, we can optimize our equipment for maximum throughput, while predicting when non-conforming material will enter the system that allows us to minimize our downtime. Simply said, we can get more volume out of our existing assets. Vision systems are helping increase direct labor efficiency by reducing training time, while improving first time quality.

As an example, these systems monitor that operators are putting the right content in the right physical location on the harness. If they make a mistake, the system immediately notifies them, which gives them performance feedback, but prompts them to correct the product immediately. In the past, that non-conformance would have only been caught at end of line inspection. Now we're saving the time and money with reworking product and cleaning the ribbing process by utilization of these systems. So we're seeing a dramatic increase in first time capability of our operators and we're seeing a very big decrease in the cost of non-conforming products entering the system.

So today, we've seen how our unique portfolio and capabilities have positioned Signal and Power Solutions to be the partner of choice for customers. Our full system solutions, supported by our world-class manufacturing expertise, provides our customers with the size, weight and total systems' advantages they're looking for anywhere in the world. Signal and Power Solutions is perfectly positioned to benefit from the transition we see and the industries we serve.

So you've heard Matt talk about the exciting opportunities in AS&UX. I've taken you through how Signal and Power Solutions is positioned to take advantage of the changing demands in electrical architecture. And I'm going to hand it off to Christian Schäfer to give you a deeper dive into the technical benefits of Aptiv's Smart Vehicle Architecture™.

Thank you.

SMART VEHICLE ARCHITECTURE™ HARDWARE

Christian Schäfer | Vice President Advanced Vehicle Architecture

Thank you, Bill. Good morning, ladies and gentlemen. I am leading the Advanced Vehicle Architecture team as part of the CTO office to ensure we leverage the full breadth of our portfolio to deliver optimized solutions for our customers.

When we originally conceived of SVA™ back in 2017, we knew already the challenges our customers would eventually face and formulated several objectives to address them.

First, simplify complexity and reduce interdependencies between the many different electronic controlled units currently required to enable functionality. Second, unite diverse applications to unlock new software enabled functionality, and deliver better lifecycle management. And third, empower our customers to fully control the software that defines the User Experience of their vehicle.

Let me now explain how we achieve this from an architecture standpoint. Our SVA™ design philosophy centers around three fundamental principles that differentiate it from legacy architectures. First, you need to abstract software from hardware due to their unequal development cycles. Second, you have to separate input and output from compute to improve the scalability and to reduce the complexity at the same time. That means we take all the physical and logical connections to the sensors and the devices and place them into a zonal controller. This also enables us to a path to affordable redundancy on all three layers of the hardware stack, power, data and compute for automated vehicles, Level 3 and above. And thirdly, when you manage the abstraction and move the IO, then you also get the ability to serverize to compute. That means we can develop high performance computers covering several domains of the vehicle and thus drastically reducing the amount of the controllers in the vehicle, which are more than 120 today. Later today, Glen and Avijit will explain how we are leveraging Wind River's proven solutions to make this serverization happen. But now I want to focus how this is physically realized in a vehicle and what hardware changes we had to make on the Brain as well as on the Nervous System as a precondition for this software-defined vehicle.

Let us have a look to the animation.

Starting with the Nervous System, we bring in the high voltage busbars, which sit directly on the battery. As Bill explained before, due to their flat profile and rigid nature, it makes it very easy to package them. Next is our patented Dock and Lock™ system. The dock itself can be assembled to the floor of the vehicle where it provides a base and then in a top-down feature all the other elements of the architecture are installed robotically. The first of these elements is our unified power and high-speed data backbone. This is common throughout every vehicle and only varies in length to support the different derivatives of a vehicle platform. This is highly beneficial for all of our OEMs as they are looking to scale the architecture across their global vehicle portfolio.

Then, we add our central compute cluster from high performance compute, Matt discussed earlier, and our zonal controllers. We have the Open Server Platform, running the complex software required for advanced safety and In-Cabin user experience. So, covering two different domains, as I explained before, we have the ability to dynamically share compute resources.

And we have the Central Vehicle Computer, which is the SVA™ Master. So, it controls all the body functionality, all the critical functions and the time synchronization is guaranteed by it between the zones.

And finally, it is the Central Gateway of the entire vehicle controlling the flow of the data in and out of the car, which is beneficial in terms of cybersecurity. Here I want to emphasize that with SVA™, the intelligence and processing power is located within the passenger compartment of the vehicle, not in the sensors like today. So helping to reduce the cost of accidents and therefore insurances.

Next, fully automated manufactured zone harnesses are brought in and connected to the Power Data Centers, our zonal controllers.

And finally, we layer in all the sensors and the actuators.

So, you see in summary, this optimized and up-integrated hardware architecture enables the fully electrified and software-defined vehicle. If you would like to learn more about this, please check out our Augmented Reality SVA™ Skateboard right over there in the entrance area where you can see it at full size and detail during the break.

However, what is even more impressive is how much value this creates for Aptiv as well as for our customers. Additional features and functions built onto traditional vehicle architectures have driven increased complexity and cost for the OEM in the last decades. SVA™ lowers the total cost of ownership in all three phases of a vehicle's life cycle: development, manufacturing and post-production.

Let me show you now an example from an architecture study we did with a large global OEM. The zonal approach with its compute up-integration and thus the reduced power and data connections, the reduced average wire length and 100% smart fusing reduced material needed in all areas. This resulted in a 10% lower hardware cost for the OEM, which is really impressive, but we also save in a number of areas that the customer doesn't directly pay Aptiv for, but is a real cost for them.

In design, it's all about integration and testing, where most of the OEMs cost lies today and can be drastically reduced with SVA™. Its greatly simplified approach and fully automatable design saves in areas such as weight, packaging space, part number management, floor space and labor. In manufacturing, many of these savings are directly related to material, but there's also substantial reduction in direct labor to install the architecture in the OEM plant.

To give you an example, today, a complex wire harness can easily take 10 to 12 people to install it in the vehicle, whereas breaking apart the physical complexity with zone controllers enables it to be done by only 1/10th. So one or two people. And finally, in post-production, SVA™ can eliminate costs associated with software related warranty. Avijit will talk about these savings in a little while. Really important for Aptiv is that while we help to lower the total cost of ownership for the OEM, we, at the same time, increase our content by 30% with SVA™ compute.

Now, I would like to hand it over to Glen De Vos, who will explain what comes next while these hardware architectures are being commercialized.

FUTURE PLATFORMS

Glen De Vos | Senior Vice President, Transformation & Special Programs

Good morning, everyone. It is great to be here to share with you the journey of SVA™ and the next steps that we're about to take. I'd like to start with how we see the car evolving as we approach 2030. As you heard Kevin emphasize, the megatrends of Safe, Green and Connected continue to accelerate and are shaping our industry. As a result, we strongly believe that the vehicle of 2030 will be highly automated on the path to full autonomy, fully electrified, and software-defined with most critical functions and differentiating features powered by software. But the 2030 vehicle will not just be defined by the adoption of these technologies. Autonomy, electrification and software are also critical to the new business models of our OEM customers as well as Aptiv. To achieve these ambitious goals that you see on the chart. What do we need to do?

Well, OEMs need advanced ADAS solutions that not only make the cars safer, but also the driving experience better and unlock high margin, recurring software subscription revenues. They need EVs to not only have the range, have the charge times and have the performance at the end users want, they need EVs that would be much more profitable. And they need software to not just drive new features that can continuously be updated, but that can also generate tens of billions of high margin revenue by 2030.

Now, Matt and Bill talked about how Aptiv is driving the commercialization of SVA™, moving from the hardware-defined legacy age basically to the software-defined and electrified vehicle. We'll, now focus on how Aptiv is leading the next phase of vehicle architecture development: the cloud-native vehicle as a part of the intelligent edge. This next step in Smart Vehicle Architecture™ and it will address the challenges our industry faces, the goals we just talked about, and it will provide the solutions for performance, cost and revenue.

But first, let's talk about what's in the way. This vision cannot be achieved unless our industry fundamentally changes the current approach to software architecture development and lifecycle management.

Why do we believe that? Well, consider the example shown on the left side of the slide. As Kevin mentioned, 60 programs, 20 different OEMs. We went back and did a very detailed study of our experience with those OEMs and those programs. The chart on the left represents the cost profile for developing, launching and supporting a highly complex software system over multiple generations. Multiple generations where the features continue to evolve. This is based on actual data verified with the OEMs and reflects the traditional approach to software development.

As you can clearly see, the costs increase exponentially over time with the introduction of each subsequent new generation. There are a number of things that are driving that. First, the legacy distributed and then domain hardware architectures were used for these and they did not provide the scalability as new features were added to the platform. It also means that the software was tightly coupled to that underlying hardware.

Second, the application software, the features in the software architecture, was monolithic, meaning it was a large block. It was laced with complex interdependencies that basically slowed down the development, the release, the testing processes, which means you couldn't really update or release the software on a frequent basis. It took a long time to get to those releases.

And third, all of this was exacerbated by the lack of a modern software development and lifecycle management tools. So combining these things results in a significant amount of inefficiencies in developing and managing this high complexity software. It limits the software reuse as well from one generation to the next as that underlying hardware evolves. In fact, what you end up having to do is basically start over with each new generation of hardware over the life of that platform. And once the vehicles are on the road, the inability to provide frequent updates inhibits our ability to one, introduce new features, but to also proactively address issues in the field, which leads to rapidly growing costs associated with software warranty.

In summary, the current approach that's being used is simply not sustainable, and it will prevent our customers from achieving those ambitious goals that we talked about on the prior slide. So what is needed to make that 2030 vision possible? Well, we need to completely transform how software is architected, developed and managed. And there are four key innovations that will make this happen, and it starts with the foundation.

I won't repeat everything Bill, Christian and Matt just shared. But I do want to stress that Aptiv's Advanced Electrification Solutions combined with their implementation of SVA™, really sets the stage, it means the physical architecture of the vehicle is ready to take this next step to a cloud-native, fully optimized modern software architecture. So we're at that point where we can now move forward.

Next, the high performance compute in the vehicle needs to be truly serverized, capable of using multiple mixed-criticality domains on a single compute device. It also needs to be powered by a modern software platform that enables a microservices based approach to the features and functions that the software delivers, and to do that across all levels of safety. So from infotainment to body controls to ADAS, every level in the car.

Serverization is also needed to break that dependency of the software on the underlying hardware in silicon. This allows greater flexibility in managing your silicon choices, it also decouples the software development process from the hardware development process, as Christian mentioned earlier. These are two very different environments in development processes that need to be separated.

Third, all the software applications that now sit on this serverized compute need to be architected with a modern, service-oriented approach that leverages containers. You're going to hear a lot about containers today. Why is that important? Containers help us break down that large, monolithic code into isolated, testable, scalable components that allows us to manage that software, to develop it, to deploy it much, much more efficiently. And to do that across all levels in terms of types of functions, including safety critical functions like ADAS or chassis controls.

And finally, to take advantage of modern software architectures, you need a modern, unified, end-to-end DevOps platform that can enable faster and more efficient development, deployment and operation of the software across its full lifecycle.

So, let's spend a little bit time on the top three software related innovations. I'll start with Serverized Compute and Microservices. Our SVA™ architecture sees compute devices basically as mini servers which no longer need to be domain specific. They can handle multiple domains, multiple functions across those. Hosting both safety critical functions like ADAS along with non-safety critical functions like infotainment, but doing so on the same compute device. Doing so enables greater efficiencies in using the compute resources on the vehicle. So, I can have more reliability and optimized performance, all at lower total system cost.

Now, in order to achieve this serverization and the fusion of those demands, onto a single compute device, I need to do two things. First, the hardware needs to be virtualized, and second, the applications that sit on that compute device then need to be abstracted from their operating systems. By doing those, I now have fully abstracted the software from that underlying hardware. As a result, software is no longer hard wired to the underlying compute hardware. And that's critical to avoid the thing we call vendor lock-in, where you're now limited in terms of what you can do with your software because of that hardware, you're locked into that SOC supply base, or into that silicon. This gives Aptiv as well as our customers much greater control and flexibility over their compute solutions.

Now, traditional automotive operating systems and middleware does not support serverization. What we are seeing though is in other industries this happening. So the virtualization of hardware, the containerization all of this is happening in other industries. For example, in the telecom industry, we're seeing 5G network infrastructure being virtualized. And what that's doing is that means that that network operator then can use standard interfaces and APIs to really have a modern software architecture across that whole network that delivers greater performance, greater agility at a much lower total operating cost. So again, we are seeing how this can be done in adjacent industries.

In addition to serverization of compute and the support of microservices, our industry needs to fully transform automotive software features as well. We can no longer structure software as one large, monolithic code set. They need to be broken into smaller, discrete modules that communicate through those standard interfaces and can be packaged into containers. It sounds familiar to do the containerization and to use microservices and Kubernetes and these orchestration tools. And the reason it sounds familiar is because this is what is done widely across other verticals. Verticals such as online streaming platforms, e-commerce, ride-sharing networks, basically what are cloud-native environments. But to bring that cloud-native approach to automotive, it also has to adhere to our real-time and safety critical needs. And those are needs unique to the automotive industry.

However, what that drives then is an operating system requirement, an operating system that has to both support containers, but also can support safety certification to the highest levels that so that it can support things like ADAS and other safety critical functions. And then finally, as we stated earlier, a modern software architecture also requires modern software development environments and tool chains. Now, as we've learned over launching those 60 some-odd programs? Well, we know that when you look at the automotive DevOps toolchain environment, it's woefully lacking. There's a lot of gaps. There's a lot of manual tool-to-tool handoffs. There's a lot of fragmentation. What that creates is a patchwork of tools that simply are not capable of supporting full end-to-end lifecycle management of the software.

But through our experience in all of this, we know that other industries actually have those tool chains. In addition, we know that a lot of our tools are open loop. They don't provide the closed loop feedback from the field on how the software is performing in real-time to the software developers. Again, automotive tool chains tend to be open loop, we need that closed feedback. So what our industry needs is a unified end-to-end DevOps platform that makes software development faster, deployment easier and full lifecycle management with that closed loop control possible.

Now, once again, we believe that this can be done in Automotive, because we're seeing this being done in other mission-critical environments such as Aerospace and Defense. An example of this is where the Department of Defense wants to ingest data from a myriad of geo-distributed devices, analyze that data, and then return those insights back to the field in real-time through a single unified platform across all branches of the military service. It's a great example of where mission-critical, high complexity environments can deliver those services. So we know it's possible for automotive.

So we are confident that by shifting to a modern software architecture and DevOps environment will drive significant benefits to the OEMs, dramatically lowering development as well as ongoing operational costs. And this is because we've seen it work. We've seen it work in our industry, as we showed you earlier, the left side represents one of the traditional and real costs associated with software development today. In contrast on the right, we show the clear benefits of a modern approach deployed by another OEM where up-integration and serverized compute was deployed from the beginning, software was abstracted from the underlying hardware, software applications followed a modern architecture and an edge-to-cloud infrastructure was built for data analytics to improve simulation and testing as well as OTA updates.

The result, as you can see, overall development and support costs are far lower, while updates and new feature releases are much more frequent and continuous, which also then unlocks the potential recurring revenue models that we're talking about. So we discussed the four key areas of innovation that we at Aptiv firmly believe are required to move to that 2030 vision of a highly automated, electrified and software-defined vehicle. Each with a clear series of key technology building blocks, and each that have a clear market precedent that shows that it is in fact possible in real-time, safety critical industries such as automotive. And through Wind River, together with Aptiv existing capabilities, we have all of the key technologies needed at every layer of the stack to enable this vision.

Building on the foundation of our advance electrification solutions and SVA™, we have advanced compute solutions that fuse mixed-critical domains. We have the Wind River Software Platform that supports containerization at the edge at all levels of safety. Aptiv software and features that are shifting to that surface-based architecture; Wind River Studio, the industry's first cloud-native and DevOps environment. And these solutions with Wind River, these are solutions with Wind River that have already been deployed and continue to be deployed in other industries, including telecom and aerospace and defense. Aptiv with Wind River is in a position to lead this industry to transformation, enabling the software-defined vehicle, driving significant savings to our customers and unlocking those new business models.

In summary, the future of mobility will be electrified and cloud-native. The digitization and electrification transformation that has started in other industries can now take place in automotive. Together with Wind River, we are able to take the next step in that SVA™ journey to the cloud-native vehicle on the intelligent edge.

So to tell us more about Wind River's portfolio and how they are helping us transform automotive software, it is my pleasure to introduce Avijit Sinha, Chief Product Officer of Wind River.

NEXT-GEN SOFTWARE ARCHITECTURE & TOOLS

Avijit Sinha | Chief Product Officer, Wind River

Well, thank you very much, Glen. And hi, everyone. I'm glad to be here to talk to you about our next generation architecture and tools for software-defined vehicles. So let's jump in a little bit about Wind River to start with.

As Kevin said, Wind River is a global leader today in providing software solutions for intelligent connected systems. Our technology enables the development, deployment, operation, and servicing of mission critical systems. Today, our customers are across multiple industries. Aerospace & Defense, Telecommunications, Industrial, Medical, and Automotive. We estimate our total addressable market to be about $21 billion by 2030. Wind River is now bringing our expertise in mission critical systems to the automotive industry to enable them to build the software-defined vehicle.

At the core of our offering, as Glen said, is Wind River Studio, the industry's first cloud-native DevOps platform for building software for the automotive industry. It accelerates software development, streamlines the deployment of that software across a fleet of vehicles, and enables the OEMs and the software developers to manage and monitor that software across the lifecycle of those vehicles. And all of these capabilities and functions are offered to them through a unified platform that is accessible to the automotive developer through a single pane of glass.

A single pane of glass is essentially the ability for developers to access all of their tools, all of their data, and all of the system and hardware through one consolidated view. This significantly improves productivity for developers because today they have to go from workstation to workstation, system to system, hardware to hardware and data repository to data repository to get access to the tools and the data they need to build the software they must build. And therefore this significantly reduces the cost of software development for automotive OEMs and accelerates time to market.

So, let's take a deep dive into Wind River's portfolios of product.

It's essentially composed of two things. Wind River Studio that we just talked about briefly and the Wind River Software Platform. Wind River Studio is a sort of solution that enables automotive developers to accelerate software development and to manage the software over the lifecycle of those vehicles.

Pipeline manager and Gallery together enable geographically distributed teams of developers to collaborate in real time and to be able to develop that software simultaneously, but independently using the tools of their choice.

Our test automation framework automates testing and enables developers to get full traceability across the entire development lifecycle.

Our system simulation and virtual lab capabilities emulate hardware for software testing, thereby enabling developers to be able to develop software much ahead of hardware availability. And this significantly improves their ability to accelerate software development because they don't have to rely on supply chain constraints or chip availability or hardware availability to start developing software.

Once they've developed that software, they now have the ability to deploy that software across a fleet of vehicles over-the-air using Wind River Conductor. And once the software is on the vehicle, data can then be gathered from the vehicle in the cloud through a digital feedback loop where the developers can use that data to build digital twins, which can then help them analyze the functionality of the vehicle and help improve that over its lifecycle.

Equally important is the Wind River Software Platform, which is a suite of product that essentially brings the cloud-native architecture into the software-defined vehicle. VxWorks, as Glen and Matt mentioned, is the only real time operating system in the industry that supports containers and is certified to safety level ASIL-D, the highest safety levels are. Our Helix Hypervisor allows mixed criticality applications to run on the same hardware, which as Glen mentioned enables the serverization of compute. Windows Linux is an open source distribution of the Yocto Linux that a lot of customers across the world use today to build mission critical embedded systems. And Aptiv technologies and middleware enable the decoupling of software and hardware by abstracting hardware from software. This significantly helps OEMs because it enables them to reuse a lot of their existing software applications.

So now let's take a look at how all of these technologies come together to enable the future of software-defined vehicles. So as Glen mentioned a vehicle today is home to a complex system of software and hardware, which is tightly coupled. Therefore, applications like ADAS and user experience applications have to be run on separate compute systems, which is highly inefficient. Now Aptiv and Wind River are simplifying this architecture by introducing a mixed criticality hypervisor that then enables the fusion of various domain controllers into one single high performance compute instance, which essentially transforms the vehicle into an intelligent edge that can connect to the cloud.

Now let's take a look at how software development happens in the automotive industry and how we can make it dramatically better. So as Glen said today, automotive software is a monolithic block of code because there are a number of components that are integrally linked through a complex web of dependencies. What this does is this makes it very difficult for developers to change the code, to evolve the code, and even produce updates that can be sent to the vehicles over-the-air. So this makes code changes very costly.

Now to demonstrate this, let's take a look at one feature, in this particular case, the adaptive cruise control capability. The software development team that's working on this looks at the code and finds there a couple of dependencies. But as they study the code base, they find that through the entire code base, there's a cascading set of dependencies that need to be changed in order to make the software work properly. But this leads the software development team eventually to one of two conclusions, either they must rewrite the entire code base or start from scratch. Either way, it results in significant engineering costs and eventually cost overruns. So this particular pattern is easy to address through a cloud-native architecture pattern called containerization. With containers, software can be made less dependent on each other. You can essentially decouple it and modularize the software and package it into self-contained dependencies. This not only removes dependencies, but makes it possible to independently and in an isolated manner update the software without impacting the rest of the subsystems. This reduces architectural complexity and development time as well.

Now it's important to note that safety critical applications like ADAS that rely on a real-time operating system can benefit significantly from containerization. But today, there are no operating systems that support containerization. And as mentioned earlier, VxWorks is the industry's first and only real-time operating system that supports containerization and is certified to the highest levels ASIL-D in the automotive industry.

So now going back to the example of the ACC, you can see here that now the ACC module can be updated without impacting the rest of the subsystems in the vehicle. This significantly simplifies software development, the testing and integration of it, and eventually the updates as well. Now within each container, it is further possible to modularize the code by making each function into a service in itself that then can enable other services to communicate with it through simple APIs or Application Programming Interfaces. Now, each function now can be separately updated. So this makes software updates even more granular, which eventually results in smaller size updates, which reduces the data transmission and data storage costs, which has today been the major impediment for automotive companies to roll out OTA services globally.

Now let's take a look at how the Wind River platform in the vehicle works in conjunction with Wind River Studio in the cloud. This essentially allows the vehicle, which is now an intelligent edge, to connect with the intelligent cloud. As we saw, the mixed criticality hypervisor and container has several benefits. But one of the additional benefits this technology pattern provides is that it can be scaled across the different subsystems in the vehicle, which then makes it possible for developers to manage the software through Kubernetes-based orchestration, not just within the vehicle, but across a fleet of vehicles as well. And Kubernetes, by the way, is the same technology that hyperscale cloud service providers like Google and Amazon and Microsoft use to roll out highly available cloud services globally. And as Glen mentioned, Wind River also enables telecommunication companies to use the same technology pattern and the same technology stack to roll out 5G services globally.

So now the vehicle can connect into the 5G network and from there into the regional data center and from there into the cloud to Wind River Studio. What that enables then is the vehicle to be able to communicate with other intelligent objects in the network, such as other vehicles, traffic lights and parking meters. This truly enables V2X, or vehicle to anything communication.

Now, once the vehicle is in this connected infrastructure, it could start sending data over the digital feedback loop into the cloud, where developers can use that data to build digital twins of the vehicles, to simulate the functionality of the vehicle, and then to identify issues and areas of improvement in the software. This drives down costs of software development and maintenance and eventually warranty costs as well.

So once developers identify issues in the code, they can easily and proactively fix these issues, thereby reducing the reason for recalls and impacting positively impacting warranty costs as well. And developers are easily able to push that software out to a fleet of vehicles and all of this is possible for them to do through a single plane of glass experience in Wind River Studio in the cloud. So our end-to-end software platform will lead to significant savings across the development, deployment and operation of a fleet of software-defined vehicles.

And we've found that during the development phase, the reusability of software itself can help OEMs realize about 20% in cost savings. And with OTA updates and the modularization and containerization of software, it's possible for them to realize about 35% cost savings, given the data transmission costs and data storage costs go down in post-production with digital feedback loop and vehicle twins, and the ability for developers to proactively fix issues and mitigate recalls. We think they can release about 25% of cost savings in warranties.

So in aggregate, all of this can result in about 25% cost savings in a typical software development program for automotive OEMs typical software development program for automotive OEMs, which can result in about hundreds of millions of savings for the automotive companies as they staff up software development teams and build software for the software-defined vehicle. And it also provides OEMs a solid foundation on which to unlock future business value.

So given the horizontal nature of our platform, the benefits of our solutions are things that we can scale across industries for customers. who want to adopt this technology pattern to drive transformation in their software-driven future. Now, as an example, in the telecommunications sector, Verizon in the US is using Wind River Studio across 25,000 sites to provide 5G services in the market. Dell is using Wind River Studio to build their infrastructure blocks, which they then provide to telecommunication companies in order to enable them to roll out 5G services. And Elisa in Europe is the first telco to have rolled out a fully distributed and fully automated 5G service using Wind River Studio. Similarly, Vodafone in the UK has rolled out Europe's first commercial ORAN-based 5G service using Wind River Studio.

And the reason Wind River Studio is so appealing to the telecommunications sector, as Glen said, is because it enables them to move away from vendor lock-in. Telecommunication companies thus far have been locked into vendors, incumbents who've provided them a vertically integrated stack, but they have no freedom or flexibility to choose providers in that stack. Now, with Wind River Studio, they can choose the right silicon providers, the right hardware provider, the right middleware provider, and the right applications to build the next-generation 5G networks. And this particular value, the flexibility, the modularity, the independence to choose is something that customers across industries want. And therefore, we are seeing tremendous interest in that regard.

So now in the aerospace and defense space and our customers like NASA, Boeing, Airbus, Northrop Grumman, Lockheed Martin, Raytheon Technologies, Rockwell Collins, they've all been using Wind River technologies and products like VxWorks and HVP for decades to build mission critical systems that operate in outer space, in air, on land, and on the sea as well. So as an example, every generation of the Mars rover that's on Planet Mars runs on VxWorks. And the Jason Webb telescope that's out in outer space taking pictures of the birth of the universe runs on the VxWorks. So we are the market leader in providing safety critical software for customers who want to build mission critical systems. And I'm glad to share today, as Matt mentioned, that in automotive, leading Chinese OEM, Geely, has chosen Wind River software platform for their next-generation L2+ program.

So we are seeing tremendous interest in adoption of our technology and products across industries by customers who want to embrace these cloud-native technologies to build their software-defined products. And much of what you saw today is available today to our customers and is being used and the fullness of what I talk to you about today is going to be available to our customers over the coming months and the near future. So we are very excited to bring all of this innovation and technology to the automotive industry so as to enable them to build their software-defined vehicles. And we are looking forward to working with our Aptiv colleagues to enable the automotive industry to truly realize the Safe, Connected and Green future of mobility.

Thank you very much for your time. I'll invite Josie Archer to come and talk to you about why Aptiv is the partner of choice for our customers.

PARTNER OF CHOICE

Josie Archer | Senior Vice President, Global Sales

[CES 2023 Highlight Reel Video]

Good morning everyone, welcome to Boston. As you just saw, CES 2023 was a fantastic event for Aptiv, with over 400 customers joining us in our pavilion, where we brought to life the software-defined vehicle, showcasing Aptiv's unique full-stack capabilities.

We debuted Wind River's Software platform, fully integrated with a vehicle powered by SVA™. We showcased our turnkey Gen 6 ADAS platform, including a public road demonstration with our customers in the vehicle. And through our SVA™ demonstrator, enabled customers to see first-hand, how these advanced architectures come together, reducing complexity and lowering overall costs.

Customer engagements continued post CES, with over two dozen follow up visits since early January. As Kevin mentioned, my name is Josie Archer, Senior Vice President of Global Sales. I'm very excited to be here with you today!

By way of background, I came to Aptiv 6 years ago, after 25 years working for several Tier One large technology providers to the automotive industry. I knew then that this was the team that I wanted to be part of. We have the right strategy, the right portfolio and the right people!

And as I will discuss shortly, this is reflected in our new business awards which exceeded $74 billion since the beginning of January 2020.

Today you've heard a lot about the latest industry trends that are enabling the software-defined vehicle including accelerated up-integration, increased software content, customer demand for more supplier resiliency, and new approaches to accelerate product development. And this is how our strengths are aligned to address those trends and our customer needs. We are the only full system solutions provider, with both the Brain and Nervous System, enabling us to create fully optimized solutions that reduce cost. We have the best technologies and capabilities with experience in over 60 complex, software-driven programs and a strong track record of execution.

For the past few years, we've managed significant supply chain disruptions and kept our customers connected, to the point that they consider us an industry benchmark, helping us win new business!

And we have a customer approach that relies on early engagement and frequent customer interactions that I will discuss in a moment. As we look into the future and how to continue to best position Aptiv, we are continuing to build on those strengths! Today, I will share with you our account-based management approach, which reflects our One Aptiv cross-functional pursuit strategy. This aligns our organization and deepens our customer intimacy, cementing our position as a partner of choice.

As part of the overall Aptiv strategy that Kevin discussed, we develop and execute very focused customer strategies that address the needs of our customers. These strategies are tailored to specific technologies and vehicle platforms where Aptiv can drive the most value for our customers and highest return for Aptiv. Let me take you through how this is actually done.

As a recognized thought leader in the industry, we take the first step engaging our customers with targeted marketing campaigns, well ahead of the start of any formal sourcing process. This allows us to identify pursuits earlier and position ourselves for success with these key targeted programs.

We then deploy our One Aptiv team approach bringing cross functional experts together allowing us to work closely with our customers and sell solutions across both the Brain and Nervous System of the vehicle. This team is led by a leader who has full end-to-end responsibility for the customer relationship and key pursuits. We also regularly measure our effectiveness through our Net Promotor System, a data driven approach that allows customers to highlight our successes but more importantly identify areas where we have opportunity to improve, allowing us to continuously strengthen our business models. As mentioned earlier, our focus on supply chain resiliency has allowed us to keep our customers connected and has been instrumental in our commercial success! This model deepens our customer relationship and best positions us for future business awards on new programs.

Now I'd like to share a real example of how this works, the One Aptiv approach in action. In 2017, we started to socialize SVA™ to a large European OEM. After months of discussions with key technology groups within this customer, we agreed on a proof of concept to optimize the full vehicle architecture, how best to optimize the signal and power distribution, along with the data and compute.

This then kicked-off our first Advanced Development Program in 2019, taking our SVA™ approach and validating the solution. We optimized the full vehicle architecture and created new solutions such as zone controllers and modular connectors, that are at the intersection of the Brain and the Nervous System.

Through the process we were able to prove to the customer that we could solve their greatest challenges. This Advanced Development Program translated into significant SVA™ business awards with this particular customer.

Naturally, the success of this approach led to additional engagements at the most senior levels to further explore opportunities around the software stack.

This is where we are today! And this is where Wind River's solutions and team come into play!

As Glen and Avijit discussed, Wind River's solutions will increase the efficiency related to software development and enable full lifecycle management, allowing the car to stay current supporting new business models.

We are excited for the next step of this architecture journey and the significant opportunities it will unlock for Aptiv and our customers!

As you can see, Aptiv invests a lot in our commercial strategies and our new business awards validate our industry leading portfolio and commercial approach. We achieved $32 billion in bookings last year, and have a clear line of sight to roughly $32 billion of bookings awards for 2023, underscoring Aptiv as a trust technology partner.

This is an exciting time for Aptiv, and I know the best is still yet to come. Having the right strategy, the right products, and the right people has truly positioned Aptiv to win across our portfolio today and well into the future.

SUSTAINABILITY IN MOTION

Obed Louissaint | Senior Vice President & Chief People Officer

[Sustainability in Motion]

Good afternoon. It's a pleasure to be here with you today. So my name is Obed, and I'm going to be co-presenting the topic of sustainability with my colleague, our Chief Legal and Compliance Officer as well as our Board Secretary.

It's our pleasure to be talking to you about the four pillars of our sustainability narrative that is focused on People, focus on Products, Planet and Platform. We firmly believe, because we're driven by impact, that when we take care of our people they'll build products that are better and also more sustainable for our planet, all under the governance of an enduring ethical platform. So, let's jump into the detail.

Firstly, Aptiv is an organization of 200,000 persons strong, you heard that from Kevin earlier. You've also heard of our focus on making sure that we attract and develop key skills, the right skills for the right time in this common era. Also ensuring that we put them into the best locations, always thinking about what's the skills depth that we need and then where the places that's going to give it to us at the most optimal costs. We're keenly focused on diversity, equity and inclusion because we know that is a differentiator and innovator for us. Health and safety is our number one priority because we know of the impact that it has on engagement and productivity. And then lastly, we're super concerned about making sure that we're giving back in the communities in which we live and work.

Let's talk a little bit more about looking at the capabilities and the people and the skills in the right places. As you've heard from my colleagues earlier today, from Bill and from Matt particularly, and then also Kevin, as we've been modernizing our capabilities and skills, we've been refining the sources of talent that we're going to. So today, we're looking at a discrete set of organizations and institutions, which is going to provide us the more modern skills and software that we're looking for, everything from AI and machine learning to software automation to computer vision and DevOps. So ensuring that we're transforming the skills and attracting them from the outside.

Additionally, we are partnering with the right universities and not just the universities, but within the universities, those specific colleges and those specific programs near our particular sites, and ensuring that we get the best engineering talent around the world.

And we are also partnering with the right content experts as well as the universities to ensure that we're up-skilling the capabilities of the individuals that are within the organization already. That's why one of the reasons that we can say that across all of our key openings, that we're able to fill those, about 50% of them, through internal placement.

We're also being able to say in this competitive marketplace, that to each one of our requisitions we're able to get about 36 candidates helping us to select the very best and the brightest in order to drive the transformation from a skills perspective.

So as we look ahead, we do see and continue to see a competitive landscape for talent. But we are confident, based on our better-than-market attrition, as well as the talent inflows that we've been able to bring in, a positive talent inflow, from our talent competitors, that we're differentiated based on our culture to be able to build and retain the skills that we need for the future. So in saying that, let's talk about what makes Aptiv's culture distinct and unique.

Firstly, it is values driven. So when we think about the core of who we are, it is important that purpose-driven or value-driven organizations. Why is that important to us? To you? It is because organizations that are purpose led, a strong culture, that are enduring outperform their competitors by 3x. They also bode higher customer satisfaction and also employee engagement. So having our culture defined by a set of foundational values then expressed through behaviors being tightly aligned with our vision and mission is critically important to our outperformance and creating a culture of operational excellence and of innovation.

And it is not about just having a set of words on the walls, but it's about ensuring that all of our people processes are keenly aligned to the culture and the values that we have. That's the way in which we select. That's the way in which we develop. That's also the way in which we recognize.

So I spoke to two key aspects of our culture that drives innovation as well as operational excellence. Let's double click on that for a moment and then very specifically look at what fuels innovation, and that's diversity, equity and inclusion.

We're pleased and we're proud of our particular progress and the fact that we're 50% women. We have gender parity around our total workforce. While we're proud of the progress that we've made, we do know that there's still work. And then when we look at each level of driving that parity, that's where our work will continue.

Additionally, when we look at very specifically in the US, we're proud to say across all of our demographics that we continue to increase today, being 43% US minority, a strong starting position for growing our workforce diversity.

But it's not enough for us to look different if we sound the same. So it is not just representation alone, ensuring that we are engaging and that's when we get the real innovation out of diversity. It is when we have genuine inclusion. So ensuring that we're engaging our workforce, also using our horsepower with our suppliers to help to diversify our supplier market.

So we know that diversity fuels innovation, so next key steps are around inclusion.

Now, the next area that I wanted to spend a couple of moments on was around well-being and in taking care of our workforce. And we know that well-being drives both engagement and productivity. So yes, we care about our people and our health and safety is our number one priority. However, we're also keenly aware that a couple of lost days of work has an impact on the productivity and engagement of the site. And as our people work on teams, one lost day impacts the ripple impact of the team. So being able to be in a top quartile, as we have for the last decade, has enabled us competitive advantages in being able to manage increasing labor costs. So it becomes a key differentiator for us.

We have several key processes which we look at every day to raise the standard or raise the bar on health and safety. So very particularly, we have things just called like the look around process where when we have an incident in one particular plant, we look at how could that remedy then be applied to all of our plants, thus always gaining the benefit and the knowledge of how do we grow and be a more safe and productive environment. And it is one of the things that engage our people.

Now, that's a bit on the people part. Great people build great products and ensure great quality in our products. Let me share with you one of these aspects, you heard earlier this morning, we talk quite specifically around Safe, Green, Connected mobility. And then when we think about our people and how committed they are to providing safer roads for your families and mine, it is one of the things that we're able to see is that on the roads themselves today, with the 40 million vehicles that have our capability, our technology in it have been able to reduce accidents by 25%. That's reducing injuries and saving lives. And when we think about green, since 2015, we've been able, as a result of our technology, to reduce emissions by over 100 million tons. And connected, ensuring that all of 100% of our vehicles are more connected, thus unleashing greater opportunity for our customers and the end users.

Great people who are well-trained also ensure great customer satisfaction and great quality. Of the hundreds of millions of parts that get sent to and from our suppliers and our customers every single day, there is greater than 99% accuracy. And then when we think about parts that may be rejected, out of every million shipped, less than one part rejection. So you can see that our team has earned the being a trusted partner and then we can see that from, as Josie described earlier, the $32 billion in bookings that was last year and expected for this year, but also from the customer satisfaction and the customer recognition that we've seen in the last couple of years.

I'll close this piece on products by talking about data security and in how we think about it in every single thing that we do. And it's a capability that's beginning at the beginning of the development process. And our team is proactive, well-trained and recognized for being for thinking secure first.

So with that, I've talked about people, talked about products. My colleague and friend, Kate, is going to walk you through the next two pillars of our strategy, planet and platform.

Thanks, Kate.

Kate Ramundo | Senior Vice President, Chief Legal Officer, Chief Compliance Officer & Secretary

Thanks, Obed, and good afternoon, everyone. As Obed said, I'll be focusing on the remaining two pillars of our sustainability strategy today, starting with our planet pillar, which is rooted in our commitment to reduce our environmental footprint and deliver products and services that make the world greener.

Specifically, we prioritize opportunities to reduce carbon emissions, improve energy efficiency, increase our recyclability and reduce our water consumption. I'll now preview some of our 2022 achievements, which we will further detail in our upcoming Sustainability Report.

As part of its mission to make the world greener, Aptiv is committed to becoming carbon neutral by 2040. To help us achieve this goal, we join the Science-Based Targets initiatives, and we're utilizing that framework to help us set appropriate targets for Aptiv.

Since 2019, we've reduced our Scopes 1 and 2 emissions by 27%, already exceeding our 2025 target. The next phase in our carbon neutrality journey is to convert all Aptiv facilities to 100% renewable energy sources by 2030. And longer-term, we'll be focusing on eliminating all other indirect carbon emissions and selling carbon-neutral products by 2039 by continuing to actively engage with our customers and their journey to carbon neutrality.

As I mentioned earlier, key to our planet pillar is our focus on increasing our recyclability and reducing our water consumption. From a waste disposal perspective, over the years, we've ensured that we have robust systems that enable us to minimize and properly manage the waste from our manufacturing sites which has resulted in a 35% reduction in that waste since 2011 and a 94% recyclability rate in 2022.

In terms of water consumption, while Aptiv's operations are not particularly water-intensive, we are aligned with best-in-class practices for water management. More specifically, we're committed to reducing water consumption in high-risk areas, and we've achieved this through impactful projects such as those that are improving water quality and consumption rates in countries like India, Brazil and Mexico.

I'd now like to turn to our platform pillar, which is grounded in the foundational principle of Aptiv's values to always do the right thing, the right way. More specifically, we're keenly focused on fostering a culture of compliance throughout the organization and engaging with our suppliers to drive sustainability throughout our supply chain.

I'll now zoom in on three key elements of our compliance program: training, engagement and risk assessment. From a training perspective, we teach our employees and others how to do the right thing the right way by deploying impactful training in various formats. We also prioritize engagement to help foster a culture that values open dialogue about compliance at all levels of the organization, and instills confidence in our people and others that their voices are being heard. Lastly, another critical feature of our compliance program is risk assessment, which we use to drive the prioritization of our compliance program. And we're proud that our focus on compliance has resulted in Aptiv being named one of the world's most ethical companies by the Ethisphere Institute for 10 years in a row.

Our commitment to compliance, though, doesn't stop with our employees, but also extends to our supply chain. As part of our supply chain compliance program, we ensure that our key suppliers are affirming their alignment with our Code of Conduct for Business Partners, which outlines Aptiv's values and priorities, including our fundamental ESG principles. We also engage with our supply base to underscore these ESG principles in other ways, for example, by supplying them with our sustainability training.

Further, during our selection of suppliers and throughout our relationship with them, we screen and monitor for various supply chain risks. And more recently, we began utilizing the digital twin mapping of our supply chain, which helps us analyze risks real-time so we can address them and prioritize them appropriately.

From a governance perspective, we believe an experienced, diverse and engaged board is a bedrock of good governance and corporate responsibility. And at Aptiv, we're fortunate to have a group of directors that's all of these things. Similar to the focus that Obed talked about earlier in terms of workplace diversity, having a diverse board is a key priority for Aptiv. It's essential that our directors have a diverse set of perspectives, skills, experiences and background, and importantly from a gender, race and ethnicity perspective, our board is 40% diverse.

Additionally, as the industry evolves and our business strategies evolve, so do the required skills and experiences of our directors. And the board has reacted to these evolving needs by ensuring it's well-balanced between seasoned directors who have a deep understanding of our business, and directors who have joined the board more recently and bring new and different skills and perspectives. Meeting our sustainability commitments also requires good governance and oversight of our Sustainability program throughout the organization.

Starting from the top, our board of directors oversees our sustainability strategy and has delegated oversight of our ESG programs to the Nominating and Governance Committee. At the management level, our sustainability leadership is comprised of a good cross-functional group of executives who oversee our overall Sustainability program and champion our specific initiatives.

And importantly, we have a centralized ESG function that focuses on operationalizing our Sustainability program, implementing effective internal controls and processes and ensuring accurate and robust reporting in alignment with evolving SEC, EU and other standards. And foundationally, our Sustainability program is informed by our ongoing dialog with all of our key stakeholders.

In conclusion, we are committed to building a strong, sustainable business that delivers long-term value to all of our stakeholders. As we evolve our Sustainability program, we continue to be recognized by many important ESG rating agencies, which helps confirm that we're on the right path. We believe that our long-term success is directly linked to the execution of our business and sustainability strategies, which together create a competitive advantage for Aptiv all while improving the world we live in.

And with that, I'd like to hand it over to Joe to provide the financial overview.

FINANCIAL OVERVIEW

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Thank you, Kate and Obed, and welcome everyone. It's a pleasure to be back in Boston for our 2023 Investor Day. And I am very excited to share our long-term outlook with you. As we all know, a lot has happened since we were last together in 2019. But despite the global challenges faced by our industry, Aptiv's resilient business model and our portfolio of advanced technologies have placed us in a strong position as we start 2023.

The acceleration of the Safe, Green and Connected megatrends that Kevin and the team discussed gives us high confidence of maintaining and then accelerating our outperformance.

In addition, as we saw during the second half of 2022 and in our guidance for 2023, our ability to expand margins not only results from material cost recoveries and elimination of supply chain disruption costs, but also from the growth of our key product lines, as well as a positive mix shift to software-enabled, fully-electrified vehicles.

Let me start with a quick recap of the past couple of years since our 2019 Investor Day. Our growth over market significantly exceeded our original expectations of 6% to 8%, finishing at approximately 12% for the three-year period. With respect to earnings, as we've discussed extensively, operating over this time period has been very challenging and very expensive. In 2022 alone, the company incurred approximately $835 million of extraordinary costs, including COVID and supply chain disruption costs, as well as material inflation. Throughout this period, the team focused on both mitigating these costs as quickly as possible and ensuring that we continue to serve our customers.

We are confident that the relentless focus on execution and customer deliveries was well-placed and provided us the ability to recover over $500 million of annual material cost increases from our customers in 2022. Our focus on execution and delivery also kept us well-positioned to win new business. Since the beginning of 2020, we have booked $74 billion of new awards across all of our key product lines, including $32 billion in 2022. We have no doubt that our ability to execute and innovate during this very difficult time period helped advance Aptiv's position as our customers' partner of choice to solve their toughest challenges.

Finally, over the past couple of years, we continued our long-standing track record of effective capital deployment, making both meaningful organic and inorganic investments to expand our capabilities and technologies while maintaining our 2019 credit rating. With that, I'd like to leave the past three years behind us, looking forward into 2023 and well beyond.

As Kevin discussed, we are perfectly positioned to take advantage of the accelerating Safe, Green and Connected megatrends. And we have the right portfolio and the right people to execute our strategy. As we look ahead, we believe Aptiv will continue to outgrow the market and expand earnings and cash flow, while investing in the business and expanding our competitive moat. As I mentioned during our fourth quarter earnings call, we have high confidence in our planning range of growth over market of 8% to 10%.

Our revenue outlook assumes global vehicle production increasing approximately 2% over the three years, reflecting our expectations for a 1% contraction in 2023, followed by a return to growth in 2024 and 2025. We expect operating income margins to be over 14% in 2025, driven by strong flow through on our above-market growth and contributions from recent acquisitions as well as higher performance. EPS growth of more than 30%, which includes the impact of our Motional joint venture and we are targeting operating cash flow between $2.8 billion and $3 billion in 2025. And cumulative operating cash flow of over $7 billion during the three years.

As you've heard today, we have a strong track record of rigorously executing our strategy to ensure Aptiv is in the right place at the right time. Harnessing the megatrends of Safe, Green and Connected to solve our customers' toughest challenges, while creating value for our shareholders. An important part of this strategy and our success over time has been a financial framework that not only supports the strategy, but helps us drive long-term outperformance.

Our portfolio of advance technologies and our commitment to investing in those technologies and the talented people behind them, help drive industry-leading growth as our record new bookings indicate. At the same time, we ensure our revenue growth is balanced across customers, platforms and regions. We remain laser focused on our cost structure and operational performance to ensure we have the ability to expand margins while investing in the business. And we remain committed to balance capital deployment, which we believe is an important lever to value creation.

Looking ahead at our 2025 revenue outlook. As noted, we are highly confident in our growth over market range of 8% to 10% through 2025 on relatively modest global vehicle production growth. The continued outperformance will be driven by strength in our High Voltage and Active Safety product lines, which are expected to grow 30% and 25%, respectively, over the period. The contribution of over $1 billion of incremental revenues from our two most recent acquisitions, Wind River and Intercable Automotive, and continued above-market growth in our Engineered Components and Electrical Distribution businesses, both of which benefit from vehicle content growth as well as strong double-digit growth in our select adjacent markets. This growth more than offsets the impact of annual price downs returning to our historical norms of 1.5% to 1.8%, and the near-term impact of foreign exchange in 2023. The continued outperformance results in estimated 2025 revenues of $23 billion to $24 billion, reflecting adjusted growth of 11% over the period, 9% above global vehicle production.

Turning now to our operating income outlook on the next slide. Flow through on incremental volume will remain strong at approximately 30% and as we look out to 2025, we expect Aptiv to return to a more normalized operating environment, including a return to historic annual price downs, a leveling off of direct material inflation in 2023. And our outlook assumes $900 million increase in labor, economics and other costs over the period, a meaningful headwind relative to the historical levels. These costs, which reflect an increase of $1.6 billion, are offset with incremental material and manufacturing performance, as well as the forecasted elimination of COVID and supply chain disruption cost in 2024.

As both Matt and Bill discussed, we continue to focus our engineering and productivity initiatives in areas that structurally lower our costs, improve service levels, and enhance the flexibility of our business model to position the company for better through cycle performance. As a result, we expect operating income margins to exceed 14% in 2025, representing margin expansion of almost 500 basis points over that time.

Now to discuss the segments in more detail. As you've heard today, our AS&UX segment has been a long-time industry leader in driving domain consolidation for complex Active Safety and User Experience systems. The segment has grown 9% over market since 2019 and has booked over $22 billion of new business awards in the past three years. These new awards as well as the investments we have made in our software, compute and systems integration capabilities will ensure that growth continuous. We expect AS&UX to grow at 16% over the coming three years, 14 points of over market, resulting in $7.5 billion to $8 billion of revenue in 2025.

Operating income margin expansion is expected in the range of 330 basis points to 350 basis points per year, with increased earnings driven by a combination of volume growth, the benefit of customer recoveries of direct material inflation, as well as improved performance. And as you saw in Matt's presentation, we continue to expand engineering capabilities and best cost locations while increasing engineering productivity.

Diving deeper into AS&UX on the next slide. Our Active Safety business has experienced strong growth despite supply chain disruptions, and the growth will continue averaging 25% over the next three years. And User Experience, we expect content growth with In-Cabin sensing and integrated cockpit compute platforms. However, as discussed, the product transition of legacy systems will mute the growth rate in the near-term with the next-gen growth accelerating beyond 2025. Lastly, our Smart Vehicle Architecture™ concept introduced over five years ago has evolved into a standalone product line with over $4 billion of bookings in 2022 and revenues commencing in 2024. Going forward, we will capture Smart Vehicle compute along with the Wind River results in this product line.

Turning now to Signal and Power Solutions, we are focused on next generation vehicle architectures, including high-speed data and high voltage electrical distribution, enabling the optimization of fully-electrified vehicles. Our full systems approach and the acceleration of high voltage electrification has led to growth over market well in excess of our 2019 targets, averaging 13% in the last three years. Further, we are leveraging our expertise in harsh environment electronics to expand into logical adjacent markets like commercial vehicles and other industrial end markets.

Through 2025, we expect growth over market of 5% to 7%, resulting in $5.5 billion to $16 billion of revenue. Operating income margin expansion is expected in the range of 110 basis points to 130 basis points per year, with strong flow through on sales growth and performance initiatives more than offsetting labor economics and pricing headwinds.

Electrical Distribution Systems is a market leader in both low and high voltage solutions, providing the power and data infrastructure required for next-gen vehicle architectures. Engineered Components includes our automotive and industrial interconnect businesses, along with cable management and fastening solutions. Within the two business units, our High Voltage Electrification product line, which totaled $350 million of revenues in 2019, finished 2022 at $1.2 billion, growing 50% per year since 2019. The adoption of Intercable Automotive will augment this product line with their differentiated modular busbars and high voltage interconnections.

As Bill mentioned earlier, strong demand for electrified powertrains and the increased content opportunity gives us confidence that this product line will more than double in size by 2025 to $3 billion. We also continue to make meaningful investments in high voltage to widen our competitive moat with organic portfolio additions of Integrated Power Electronics and Battery Management Solutions. In summary, our Signal and Power Solutions business has the global scale and flexible cost structure to remain a market leader.

Now looking at our adjacent market opportunity in a little more detail. Over the past six years, our focus on identifying and capitalizing on opportunities for our advanced technologies and adjacent markets has added approximately $12 billion of cumulative, accretive revenues to Aptiv. Through a combination of both organic and inorganic investment, our capabilities around harsh environmental electronics and centralized compute has driven growth in commercial vehicle revenues, up 12% over the commercial vehicle market since 2019 and we've experienced growth in industrial, aerospace, defense and telecom.

And since late 2021, we've been actively growing our presence in the commercial space market, where we saw revenues grow 80% in 2022, albeit off a relatively low base. As we enter 2023, we will continue the expansion into the markets leveraging Wind River's position as a provider of complex Intelligent Edge Software Solutions to the 5G telecom market as well as aerospace and defense. As we look out to 2025, we expect adjacent market revenues to exceed $4 billion, representing almost 20% of total Aptiv revenues. And although at this level, adjacent market revenues fall short of our original 2017 aspirational goal of 25% by 2025, in part due to the higher-than-expected automotive growth, we are committed to continuing to grow and expand our adjacent market presence, both organically and inorganically.

As I noted earlier, an important element of our financial framework is ensuring we continue to invest in the business and fund that reinvestment through improvements in our operating performance and optimizing our cost structure. In addition to the flow through on incremental volumes, margin enhancement will be driven by improvements in material and manufacturing performance as a percent of sales, including strong focus on Lean manufacturing, reduce freight and logistics costs and further optimization of global footprint and continued rotation to best cost countries. Advancements in engineering tools, processes and growth and best cost engineering centers will allow us to maintain our relative engineering costs while supporting the continued advancement of our technology portfolio and supporting record customer launch activity over the coming years.

In addition to funding innovation, the savings in operating performance fund investments to improve our operational resiliency. Over the last few years, we've made investments in regionalizing both our manufacturing footprint and supply chain, as well as investing in technology and team members focused on resiliency. As of the end of 2022, approximately 90% of what we manufacture today is in region, directly aligned with our customers' regional footprint. With additional investments in the next three years, we expect that percentage to increase to 95% by 2025. With respect to our supply chain today, we source more than 75% of direct material within region, up from 2019 and we are continuing our efforts to increase our supply chain regionalization with a particular focus on electronics, a category that is currently heavily dependent upon a supply base in the Asia Pacific region.

To help manage and drive increase resiliency, we have invested in technologies that allow us to map and model our supply chain, providing greater flexibility and increased response time to dynamic changes. These investments have significantly improved our operating capabilities over the past several years and as Kevin noticed on our Q4 earnings call, they have been recognized by our customers. Over the past two years, Aptiv has been the recipient of more than 70 customer awards in service, quality and supply chain effectiveness.

Moving now to capital deployment. Our sustainable business model is enabling us to convert more income to cash, with 2025 operating cash flow expected in the range of $2.8 billion to $3 billion, and with cash flow compounding by approximately 30%, there is no shortage of attractive deployment opportunities. We will maintain a well-balanced approach to capital allocation as we continue to prioritize organic investments in the business to support our portfolio of advanced technologies and record new business awards, execute our M&A strategy and focus on transactions that enhance our scalability across both the Brain and Nervous System of the vehicle, accelerate our speed to market with relevant technologies and access new markets. And while we will continue to maintain our current financial policy as it relates to our balance sheet and leverage profile to the extent we can take advantage of market disconnects, we will be opportunistic in share buybacks and returning cash to shareholders.

As we announced on our Q4 earnings call beginning in 2023, we will re-implement our prior practice of minimum annual share repurchases to offset the dilution from incentive compensation plans approximately $100 million a year. In summary, the almost $7 billion of cumulative cash flow we plan to generate between now and 2025 will allow us to build upon our track record of disciplined capital deployment, which we believe is a major differentiator for Aptiv and an important lever for shareholder value creation. The next slide expands upon that point, outlining our capital deployment track record.

Since our IPO in 2011, our capital allocation strategy has driven significant value creation as we have invested in the business to grow our technology portfolio and support new business wins, expanded geographically into adjacent markets through M&A, and returned over $7 billion of cash to shareholders. As you see from our M&A track record, our process for identifying targets and ensuring we capture value from acquisitions is as rigorously executed as any part of our strategy. Since 2017, we have completed 20 transactions and believe we are well-positioned to remain active. The addition of Wind River also provides a platform for additional software acquisitions that will add to our capabilities.

In addition to meeting and exceeding our synergy targets, excellent acquisitions like HellermannTyton benefit from our commercial excellence initiatives and being incorporated into our full system solutions while also helping drive our adjacent market expansion. And we are confident our recent 2022 acquisitions of Wind River and Intercable Automotive will continue to build our track record of over performance.

To date, Intercable Automotive has seen an increase in new business award opportunities, with a current funnel of $7 billion, up 40% from last year. In addition, we've already started expanding their manufacturing footprint into North America, allowing Intercable to more effectively serve North American OEMs. And despite the extended period between sign and close, Wind River and Aptiv made significant progress under our prior collaboration agreement to develop a full automotive software stack. And as previously mentioned, Wind River had their first win with Geely.

The execution of our strategy will continue to drive revenue outgrowth and margin expansion into 2025. However, as the team has discussed, we see the Safe, Green and Connected megatrends accelerating as we move beyond 2025 towards the end of the decade, and Aptiv is perfectly positioned to benefit, accelerating our growth, increasing our earnings power and cash flow generation.

Post-2025, growth over market will accelerate above 10%, driven by higher levels of key technologies like electrification and Intelligent Edge Software Solutions, and our presence in key adjacent markets will continue to grow.

Further margin expansion will be driven in part by our revenue growth, but will also benefit from enhanced product mix, including the growth of recurring software revenues. And we remain confident in our ability to manage through key macro factors as we continue to invest in a business that is more resilient and laser focused on performance and execution.

Moving to the next slide and our vision for 2030, the intelligent transformation of Aptiv that is taking place today and accelerates as we get to 2025 and beyond, builds a faster growing, more diversified and sustainable business. With estimated revenues of $40 billion in 2030, comprised of the next generation of advanced technologies that drive mobility, including cloud-native software stacks that manage the full software lifecycle of the vehicle, increased levels of automation and digital transformation of the cockpit and user experience.

Connectivity that enables new business models for our customers and Aptiv. And optimized architectures that provide the power and high fidelity signal to enable automation and maximize efficiency. We expect Smart Vehicle compute and software to total approximately $6 billion of revenue in 2030, with Wind River representing more than $1.7 billion of that total. Aptiv's total software revenues in 2030, including Wind River as well as our application software included in the other AS&UX product lines is estimated to be approximately $3 billion. We estimate that approximately 60% of the software revenue will be comprised of recurring license and service fees, not directly tied to units of production.

The accelerated growth, higher volume levels, and favorable portfolio mix will help drive total Aptiv operating income margins in 2030 to more than 17%, reflecting 300 basis points of margin expansion from estimated 2025 levels, and annual free cash flow is expected to exceed $5 billion by 2030.

Before I hand the event back to Kevin for his closing remarks, let me wrap up with our investment thesis. As you can see from the slide, Aptiv is growing significantly faster than our industrial peers. In addition, revenue upside opportunities exist within our key product lines, and our strong cash flow generation with over $2 billion annual free cash flow beginning in 2025 will provide the opportunity for M&A upside to expand our capabilities and the end markets we serve. And as discussed, we will continue to strengthen our margin profile and cash flow generation. We believe this adds up to a compelling entry point at current valuation levels as the vision of the company in 2025 and 2030 increasingly comes into focus.

And with that, I'll invite Kevin back up on stage for his closing remarks. Thank you.

SUSTAINABLE VALUE CREATION

Kevin Clark | Chairman & Chief Executive Officer

thanks, Joe. As Joe said, I'll wrap up briefly before we open up for questions. Joe said the last three years have had a few challenges, but as you heard from me and you heard from the rest of the management team, we feel like we have tremendous momentum. And this is a management team that's dealt with those challenges on a day-to-day basis. And we feel that way because we've been very intentional and we've been very rigorous, continuously refining and then executing our strategy with truly laser-like focus, evolving our portfolio, as you've seen today in response to the accelerating Safe, Green and Connected megatrends and creating full systems solutions that increase performance and lower costs and address the needs of our customers and enable the path to the electrified, software-defined vehicle.

And then further strengthening our already optimized and resilient operating model that delivers for our customers and had significant operating leverage, growing sales faster than costs and converting more of our income into cash, which we can invest organically or inorganically to enhance our portfolio of advanced technologies, which again further widens our competitive moat, all which uniquely positions Aptiv to deliver value to our customers and create significant long-term value for our shareholders.

So with that, I'd like to invite the speakers up here, so that we can answer Q&A.

QUESTION & ANSWER SESSION

Jessica Kourakos | Vice President, Investor Relations & ESG

So thank you, everyone, for joining us. We are now ready to take your questions.

We'll start with people in the room. If you can wait for a microphone to come your way, if you can state your name and company affiliation before asking your question, we appreciate it. And for those of you in the remote audience, again, the microsite does accept questions, so please feel free to enter your questions there as well, we'll try to incorporate you into the Q&A process as best we can.

So with that, Kevin, we can get started.

Rod Lache | Wolfe Research

Rod Lache with Wolfe Research. Just two questions. First, just financially, it looks like you're expecting some acceleration in margin expansion beyond 2023. So this year will be 140 basis points, 150 basis points something like that. And when we looked at the bridge, there were some significant inflationary pressures that are kicking in another $300 million this year, but maybe you can just address why those sort of attenuate and you can accelerate about 200 basis points per year?

And then secondly, just thinking about this paradigm shift in product development that you were talking about, I'm curious about what automakers alternative is to that? Are competitors of yours offering different development schemes? Could there be a QNX development scheme or something like that? And could that actually come back and create more complexity for Aptiv if you have to deal with different development protocols from each of your customers?

Kevin Clark | Chairman & Chief Executive Officer

So, Rod, great question. Why don't I start with your second and then Joe will respond to your question about our outlook. Listen, having been in the industry for a long period of time and as we intentionally talked about the number of advanced software and systems integration programs we've been involved in for the last number of years, we spent the last couple of years spending a lot of time kind of identifying where the pain points were for our customers and then for Aptiv as well as, I guess, the broader supply base. And what are the underlying trends and what's necessary to address those pain points? So I would say we spent the last three years really talking to a number of players, pretty close to virtually all the players, they touch every aspect of that full tech stack that we talked about and as you talk about things like operating systems middleware, tool chain development from a traditional standpoint, there was literally no one out there who was as advanced, even close to as advanced, as where Wind River is now and as when we first entered into our commercial agreement in the aerospace, defense, telecommunications and other markets.

And importantly, those were markets that literally went through the same challenges that the automotive industry is going through today and given Wind River's 40 years of history operating in those markets that were mission-critical, that needed real-time safety certification so very comparable operating environments, Wind River was the obvious choice from an overall commercial partnership standpoint. And then when we saw the opportunities outside of auto that remain as well as inside of automotive, that's what led to ultimately our acquisition of Wind River. I think it's possible and you're never going to hear us say that someone could change their strategy, but we would tell you Wind River is years ahead of any of the traditional players that operate within that overall real-time operating system middleware space. And we spent a lot of time with those who weren't in, who aren't in that space today that you might think would have interest in it. And they lacked that experience as it relates to top mission-critical in safety certification, the needs that our industry has or the needs of the aerospace and defense industry has to operate safely.

Now, could an OEM choose a different path or take a slower path? It's possible, but I'd say we're very, very confident that the path we're on with Wind River with our software above that Wind River middleware from a hardware standpoint is actually the path the industry needs to head down to actually enable all of the battery electric vehicles and all the off-vehicle revenue opportunities that OEMs are actually talking about today.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah, Rod. I'll just go back to the margin question. You're absolutely right. It certainly builds over time. There's sort of two. You can think of them as three reasons why that happens, right? There's a couple of sort of immediate, larger upside opportunities and 2023 will benefit from the full-year impact of all the customer recoveries on the direct material inflation. We really started recovering in June or July last year when the recovery started to pick up. A few of them were retroactive, but the bulk of them we sort of lap in the first half of this year. So there's a margin step up included in the 2023 guidance. Another large item that comes off in 2024, we're estimating by then the supply chain disruption costs have effectively gone away. There are $180 million estimated in the guide for 2023. So, whether that's zero in 2024 or goes away early in the year, I think that's within the range, but that's a big number that comes out quickly.

Then two other things I would say that's sort of builds slowly overtime. One is the product lines continue to grow and as you've seen in the past, we benefit a lot from scale, right? As we get a lot of efficiencies from manufacturing and things like high voltage at scale, better throughput, stronger just manufacturing performance. Those will continue to build those product lines grow. And then the third is just we're going to start to see a better mix and I think we've experienced this over the last couple of years. I appreciate there has been times where it's been a little bit hard to see just given everything else going on and going through the P&L. But, and Bill mentioned in his prepared remarks, the vehicle today and our expectation for the vehicle tomorrow tend to be more contented, particularly if they're electric, right?

If someone's buying an electric vehicle, their benchmarks of the Model S or the Model 3, depending on their price points, they tend to get a fair amount of digitalization in the cockpit, as well as more advanced Active Safety systems than their ICE equivalent model. So that positive mix, I think, continues to flow through the P&L and grows, as we get certainly into 2025. But then beyond that.

Emmanuel Rosner | Deutsche Bank

Thank you so much. Emmanuel Rosner, Deutsche Bank. Two questions as well.

Just first following up on the margin question again, just for a little bit more detail. So when I'm looking at your walk through 2025, you have about $1.7 billion of performance and lower supply disruption. Only about $400 million of that comes this year in the 2023 guidance. So specifically within the inefficiencies, you have a meaningful acceleration of performance, which historically has been difficult to achieve in the in this kind of industry.

So can you maybe talk specifically, besides your assumption of some of these disruptions going away, what are you doing specifically on the cost structure to make it more flexible? I know you have a slide around sort of like trying to reduce the various parts of the cost structure, what is essentially in your control to make sure that you can unlock in those larger efficiencies over the next three years and mostly beyond this year?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Sure. Let me hit on a couple of and maybe I'll ask Bill to chime in on manufacturing performance. As you saw on that slide, I think there's a couple of big things that will sort of jump out. One is holding engineering expense at roughly that 6% of sales, despite the growth in revenue. That has been a focus of ours engineering efficiency ensuring not only we have the right people, but they are in the right place has been a big focus. You heard both Bill and Matt talk about continued engineering rotation of best cost countries, opening of tech centers in India. So maintaining a very consistent engineering expense, even though as sales grow considerably helps obviously.

Material performance is something else we're very focused on, right? We've come through a period of supply chain disruption and some price inflation. And we've reset with a number of suppliers. We've reset our expectations for improved performance. And quite honestly, we've exited a number of suppliers and downsized the supplier community in part from an economic perspective of how can we get better strategic relationships of people that one can participate suppliers that can participate in our growth. They see our bookings as does the rest of the world, but also that are committed to giving us some annual cost improvements. And we've been helped a little by the regionalization there, right? We've been able to expand the aperture on who we're dealing with from a supply perspective, have been able to go into the other regions and maybe look for some suppliers that meet our quality standards, meet our delivery standards, but haven't necessarily been part of our supply chain before and we're able to get some benefit from an economic perspective. Do you want to hit on just Lean in manufacturing?

Bill Presley | Senior Vice President & Chief Operating Officer, President Signal & Power Solutions

Yeah. I would say on the manufacturing side, I mean, we have a couple of monuments that we focus on. One is we're focused on increasing revenue density, specifically pushing more volume through our existing infrastructure. Good examples of that is if you looked at Europe back in 2019, we actually put more revenue through Europe now with 450,000 square feet less manufacturing space than we had.

And that comes in the form of a couple of key things. One, I said, is increasing overall equipment effectiveness when we do that and we get more throughput on our existing assets, we're actually able to clear machinery off the floor to put in more final assembly production, which is the value add that we specialize in and then by increasing our direct labor efficiency and improving our first time quality that we do that eliminates structural cost from every plant. So those are the key monuments that we focus on.

Emmanuel Rosner | Deutsche Bank

And then second question is Motional. So you didn't really have any dedicated slide to Motional. You've mentioned that in the financials around what the EPS would look with or without. What is, I guess, the strategy here, are you dedicated to this asset, when is it going to move away from being sort of essentially a drag of about $1 through EPS to something that could be contribute?

Kevin Clark | Chairman & Chief Executive Officer

Thanks for asking that question. So I talked about Motional. I think Matt talked about Motional as well. Listen, I think it's important to understand we view Motional, we've always viewed Motional as our broader Active Safety Solution, right? As a part of that overall continuum and maybe different from a number of different players that today Motional was a customer of Aptiv and quite frankly, Aptiv is a customer Motional. So we bring their technologies into that. For example, the Gen 6 solution that that I referred to, from a tech roadmap standpoint, the business continues to do extremely well. They'll be fully driverless in the Uber and Lyft networks starting in Las Vegas later this year.

So from a tech standpoint, on track. From a revenue standpoint, back in 2019, I believe we said revenue expectations were roughly $500 million in 2025 today. Our outlook would be roughly half of that, but it's still a very important part of the overall mix of assets that we have in Aptiv, and we feel like it brings value to our existing business and it provides certainly opportunities in autonomous driving from an overall traditional autonomous driving standpoint as well as autonomy and other markets.

Adam Jonas | Morgan Stanley

Thanks. Adam Jonas with Morgan Stanley. First, thank you for having us and providing all this great detail. It's really refreshing to see this level of transparency. Your message is pretty clear from the team a faster EV penetration is positive for Aptiv in terms of margins, returns, growth. At the same time,

for the majority of your OEM customers, they lose money and they're going to lose money in this segmentation for some time. IOf course, excluding Tesla. So how do you reconcile this problem? And I'm thinking, in other words, if the EV penetration is slower than expected, either due to geopolitical reasons or economic reasons, is that adverse to you achieving these targets, or is it ways to neutralize that?

Kevin Clark | Chairman & Chief Executive Officer

No, I think that's again, it's a great question. As we showed our underlying assumption as it relates to EV penetration is below what at least some of the industry benchmarks are out there. So we've taken a cautious, what we view as a cautious approach to overall market penetration. The second piece included in that, we've been very selective in terms of the OEMs that we partner with from a high voltage electrification standpoint. So our primary customers, our largest customers are Tesla, are Volkswagen, are Volvo, and few others where they have, they build dedicated EV platforms that they are taking across multiple markets and then across multiple vehicle platforms. So that it would give us a higher level of confidence in achievement of their volumes, even if there's some pressure on overall industry volumes. So that we felt like if we're making the investments, we're closely aligned to their growth profile. I think we feel like based on where we sit today, the players that we're partnering with today, that we're very prudent in terms of what we've baked into our outlook. I'm sitting here today and I certainly fully understand where there may be some concern about any sort of headwinds to overall EV adoption. We would say we've captured that in our overall outlook based on our underlying industry assumption as well as the selectivity of the OEMs and their platforms.

Adam Jonas | Morgan Stanley

Great. And just a follow-up. Just confirming that you've had a relationship with Tesla for a long time, you guys. It was a good bet for a lot of reasons, economically, information, et cetera, strategically. Just confirming that the content per vehicle there is really focused on the low voltage still that has not changed.

Kevin Clark | Chairman & Chief Executive Officer

No, we have mix of low voltage and high voltage today. I don't know, Bill, you can probably...

Bill Presley | Senior Vice President & Chief Operating Officer, President Signal & Power Solutions

Yeah. We do the high voltage in both the model Y and the model S. So, we're a strategic partner with Tesla across the board.

Adam Jonas | Morgan Stanley

Okay. I've just asked that question because when you think of just players that could have a disproportionate share of the EV market, particularly in the West, there's some scenarios where Tesla could have a very, very large chunk and so that content growth is going to be and also embedded in your forecast as well.

Kevin Clark | Chairman & Chief Executive Officer

Yeah. I would say, Adam, with Tesla, Josie talked about our account-based management approach. I would say that's a great example where we've implemented that approach and have been very strategic with the customer and have grown very significantly on existing platforms, on new platforms, and then globally. So Tesla has gone from, I don't know, five or six years ago, very little revenue for Aptiv to today I think in 2022, they are the fifth largest customer. So they're a customer where we've done a very good job partnering with Tesla and enabling them in areas where we can add value, and we found common ground. So they've been a great relationship, a great customer.

Itay Michaeli | Citi

Thanks. It's Itay Michaeli from Citi back here, just two questions. Kevin, I think in your opening remarks, you mentioned that disruption is the new normal. And I'm just curious, at a high level, when we look at the outlook for 2025, how normal is 2025 in your scenarios relative to pre-COVID, how much cushion is there for just general things gone wrong? And second question maybe for Joe, you mentioned that the software revenue in 23, I think 60% of it being tied to non-production related revenue. I'm curious, how big that could be over time. Like what's the penetration of, vehicles where you would be generating that kind of revenue in 2030 relative to the overall market opportunity.

Kevin Clark | Chairman & Chief Executive Officer

Itay, I'll start with the first and then Joe you can add to it as well, to answer the second. Listen I think there is an element of how do you think about how much is enough as it relates disruptions. Right? If you think about COVID and you think about semiconductor shortages obviously standing in front of you folks in making commitments and I think all of you know us we take our commitments very, very seriously.

We've taken a very what we feel is a realistic view on what is the supply chain look like, what is capacity look like, what is vehicle production overall look like and have arrived at an area where we feel very comfortable with the projections and the commitments that we've communicated to you today.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah, I was just on that question. I would say at some point and 2025 really is the first year. As I said, we see the disruption cost going away in 2024. We've made a lot of investments around resiliency in the business, and those investments are working. I think we always had a good handle on our supply chain. It's exponentially better at this point in terms of clarity, data, visibility, and our ability to respond. I mean, we're now responding in the day to something that maybe took four or five days a few years ago. It's been a significant improvement. Obviously, the 2030 revenue assumptions are just that their assumptions we feel like they're grounded in some good basic assumptions. Wind River historically has run at about 80% license and service fees. So that's included in our on the Wind River growth. We don't anticipate their business model changing. And then we've made some assumptions that I think are a relatively conservative around call it 10% to 15% of Aptiv's application software over time, moving from a sort of price per unit to some subscription-based.

And I think that's reflective of our thoughts on new business models. In reality up-integration as you guys have heard today where we believe up integration is going to happen, there is going to be software that it's gets up integrated into other's boxes and the pricing around that or how that's consumed economically I think it's going to be a little different. So again longer term estimate based on some conservative estimates on the Aptiv side and then a long track record of how Wind River goes to market. So we think they're really good place to start from an estimate perspective.

Luke Junk | Baird

Thanks for taking the questions Luke Junk with Baird. First question I wanted to ask about the 2025 margin bridge for AS&UX. So in the presentation, you talked about several elements that are driving that volume expansion from product mix in terms of smart vehicle compute and software and some things that you're doing internally in terms of best class countries with respect to engineering, adoption of Wind River tools.

What I'm wondering is, if we can just discuss how those pieces breakout and specifically what is under your control or the structural margin improvement that are envisioned in that outlook? And then for my follow-up, if you could just also speak to the evolution of Wind River's revenue mix, both with respect to the 2030 target you've laid out and maybe more importantly nearer term as well. Thank you.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah, let me start with that one, because it hasn't changed since the discussion last January. We're targeting around $1 billion of revenue in the 27 time period for Wind River. And we expect that to be about 30% automotive. And obviously grow from there. So I think that's probably a pretty good sort of proxy. As it relates to what's in our control, obviously, we've tried very hard to make sure things that are in our control or what are driving that forecast. So Matt spoke to the continued best cost rotation. Obviously with the more favorable mix around software and the software defined vehicle with Wind River's part of an AS&UX segment obviously we feel that's very much in control.

And I think, as we turned into 2023, those customer recoveries, it was a big number for Aptiv overall. But if you think about 75%+ of that sort of hit the AS&UX segment, it's even a much bigger total on the smaller of the two segments, right. So those are in place. We've booked about $0.5 billion, as I mentioned, in 2022. We're going to get the rollover benefit to sort of cover the first half difference between 2022 and 2023. So there's obviously execution involved. There's a lot of work involved to continue to hit those targets. But I think we've done a pretty good job of identifying things that are in our control or in the house at this point that we can leverage.

Kevin, do you have any?

Kevin Clark | Chairman & Chief Executive Officer

No, I think you covered it. Listen, I get back to the last couple of years have had some unique challenges associated with them. Supply chain is improving, although there are times that we're still dealing with choppiness. COVID situation is improving, but there are periodic outbreaks that we're dealing with. And as you look at the projections, we've given today, those are all factors that we spend a lot of time thinking about analyzing and evaluating obviously before we communicated them to folks like yourselves though.

John Murphy | Bank of America

Good morning. John Murphy, Bank of America. Thanks again for all the detail today. Just a first question, I only got two. When you think about the new products launching right now whether be on the ADAS or autonomous or EV side, we've seen a lot of companies run into sort of this boogieman of launch cost on

R&D and CapEx, because a lot of this change is very burdensome on sort of the entire systems. I mean, what gives you sort of the confidence that you're not going to run into those same issues? I mean, you haven't yet historically, so right. I mean, is there something inherently in the way you operate in the product set, the relationship with the customers or what you're going after that you haven't run into those major issues?

Kevin Clark | Chairman & Chief Executive Officer

Yeah. I think when you look at our major programs, especially those programs that are related to advanced compute and software, over the last five years we established a product organization. We have made a lot of progress on productizing those overall solutions, so much more significant levels of reuse of existing software, clear line of sight to program deliveries both from a customer standpoint as well as from a supplier standpoint and disciplined standardization that we're driving standardization in terms of what we commit to, what we deliver, and what that translates to from a platform standpoint.

So, Matt referred to that Asian OEM as an example and the solution that we're providing them. That solution is actually, I won't say it, quite a carbon copy, but it is a lift of a solution that we've actually already built for other customers and are overlaying with that particular customer with some small amount of customization to give them their look and feel that they want in their vehicle.

John Murphy | Bank of America

Okay. And then when we think about the revenue assumptions for 2030, how much of it is an adjacent markets? And when you roll into those adjacent markets who are the competitors you're running into?

I mean, is it just kind of like free rein because you're so good, you're coming from our orders and it's a great operating environment to test out in, and you can just kind of roll in and take over or you're running into some real competition there? Because it sounds like you're having a lot of success there but I got to imagine there's a pretty good competitive set there as well?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah, it's right around the 20% level. We've sort of maintained as the number has grown, obviously we don't forecast in M&A, right. So, there would be some M&A upside potentially to that number.

Listen, our focus has really been on the adjacent market side, particularly on the M&A side or on the product development of something like CV, to identify those areas where we think we're incrementally better than the existing competition or provide and this is clearly the case in CV, provide the OEs in that space with choice they did not have, right.

So, we have a competitor that at one point had 85% plus of the connector market in CV and we had a customer base there that was actively looking for alternatives, particularly around some certain technologies. So, we continue to be very selective on that and trying to pick those spots. And I think who we run against, you'd expect certainly on the connectors' side, they're the known names that you would expect. And quite honestly, we see on the CV side a couple of our automotive competitors. But again, I think there it's sort of the advanced technologies and the ability to deliver what is a smaller market from a unit perspective, to be able to deliver them a very cost effective solution, given that we've, sort of amortize the cost of developing some of those programs across our automotive volume.

Kevin Clark | Chairman & Chief Executive Officer

And so if I can add to that, John, I think it's important. Listen, we're very deliberate about everything we do there is if anything in the last three years have proven that anything there is nothing that's easy.

We don't roll in anywhere and we approach each one of these opportunities very intentionally with the great deal of knowledge of where we can be successful, what can we bring from a value standpoint. And how do we penetrate and how do we drive higher margins? So you think about Wind River. That's a business growing 20% in aerospace, defense, telecommunications in other areas. Strong growth in adjacent markets based on the solution that they provide. Two years of evaluating other players that provide similar solutions across multiple markets, no one with the containerization capability, the micro-services capability, the Wind River Studio, which is the toolchain capability, and then the ability to do real-time safety certification. Which is massively important when you think about aerospace and defense. Those are solutions that sit on the F-15 fighter. They're going to work on a passenger vehicle in the automotive market, right. And they're the exact same issues. They have to work all the time. And they're the only solution out there that enables, truly enables our customers to do what our customers are saying they need to do.

John Murphy | Bank of America

Driving in New York City is kind of like a dogfight. So yeah, so I hope it works the same way. I'm just on the margin expansion of 300 basis points there. How much of that comes from the adjacent market growth for 2025 to 2030, you have 253 basis points of margin expansion. How much of that comes from that mix?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

I'd have to get back to John with that exact number. They were it remains accretive to the overall Aptiv to SPS and Aptiv.

Colin Langan | Wells Fargo

Colin Langan, Wells Fargo not to keep repeating the same question, but the $1.7 billion you mentioned $200 million or $180 million was supply chain. You mentioned a bunch of things like mix, supply, scale. Any way to parse the other $1.5 billion out into and what are the main drivers of that performance, just things like a big bucket that would be kind of helpful to get a frame, so?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah. The volume obviously wouldn't be part of that. That's on that walk, right?

That's on a separate part. I'd go back to that and I will rattle off the numbers of my head, but I'd go back to that chart that show the improvements in the material, the manufacturing mix, that is effectively the performance that is going into that walk. As well as to your point, the $180 million coming off in 2024.

Colin Langan | Wells Fargo

Okay. And then you obviously have already have a pretty low cost labor footprint. I think you're one of the biggest employers in Mexico, if I'm not mistaken. What can you do with the wage inflation down there to kind of help offset that impact? Probably pretty, it's obviously material to, based on your guidance. Can you relocate anywhere lower cost or what can you do there?

Kevin Clark | Chairman & Chief Executive Officer

I'll start and then Bill should chime in. There are a number of things that can be done.

Clearly, moving away from the border is one. And that's one of the items that we're doing while staying in Mexico. There are areas outside of Mexico and Central America that are now attractive from an overall supply chain standpoint. So we have manufacturing operations in Honduras, for example. There are other countries as well that we're exploring that we likely move some of our operations into. And then I think, as Bill talked about in some product lines, there's the opportunity to automate some or all of that overall process.

Bill Presley | Senior Vice President & Chief Operating Officer, President Signal & Power Solutions

Yeah, I would just echo, right, what Kevin said is true. We're very deliberate on everything, deliberately where we are placing things like high voltage systems that have a higher amount of automation near the border where the inflation tends to be. And we are moving higher direct labor content into the interior where the inflation is not as prevalent. And I go back to what I said before. We're obsessed with improving direct labor efficiency and overall equipment manufacturing. Our objective is not to add more assets. Our objective is to maximize the throughput of our existing assets.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah. And I would just add, the labor inflation I think is an industry or global phenomena, there's nothing in unique to our footprint that's driving that incremental labor inflation. I think most folks will deal with it. They and we will continue to deal with it. We've been very focused over the past year and a half and will continue as we first start seeing it of setting the expectations that we have to have operational performance offsets to deal with it. Whether that's taking labor out of the plants for Bill's point or finding other levers that act as actual sort of dollar offsets.

Tom Narayan | RBC Capital Markets

Tom Narayan, RBC. Thanks for taking the questions. Some OEMs talk a big game about doing assisted driving software, taking it in-house. To what extent is that they're, maybe being naive or do you really take that as a real potential possibility? And then the second one is, you know, on the AS&UX 2025 revenue and EBIT targets, which are quite impressive, how much is that already coming from booked business?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah, sure. I can start with the last one, Tom. I mean, from an obviously dependent upon vehicle production. But as you think about sort of how we book business and how we book revenue, you're in obviously basically almost 100% for 2023, high 90s for 2024 and sort of low-ish 90s for the third year out. Now, again, it's going to change based on vehicle production. But at this point, knowing what we're building for who and when is one of the benefits of our booking cycle, right. Our bookings, they do take two or three years to convert from bookings to revenues. But once their revenues, they tend to run the program life for five to seven years.

Kevin Clark | Chairman & Chief Executive Officer

As it relates to OEMs bringing certain activities and software, others. Listen, there are you take software as an example. Clearly, there's more software content going in the car. Clearly, there are some areas where it makes sense for some OEMs to do more of that software. To not change the process for developing software, Glen showed the analysis that we had done relative to a number of programs that have been out there across various domains and what total costs have been for OEM all the way through the supply chain versus another OEM that does it significantly differently. If you in-source doing it the same way, there is not a lot of savings. There are areas like middleware like operating systems, like other areas that it makes no sense and makes zero sense for an OEM to bring in and develop their own sort of middleware or operating system for their vehicles. And you've read a lot of the OEMs have had some trouble in-sourcing some of the activities.

Mark Delaney | Goldman Sachs

Thank you. It's Mark Delaney from Goldman Sachs. Appreciate the presentation today. After expecting growth of a market to be 8% to 10% through 2025 and then you spoke about it accelerating to 10% growth of a market starting in 2026 through 2030. Can you elaborate a bit more and how much visibility you have into that acceleration in growth over market? And perhaps how much has already been secured with bookings and especially in software, which is a big part of that 2030 margin target?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah. I think as you get out past 2025 we do see it accelerating above 10%. That's when the SVA™ revenues start to kick in. So we obviously get some SVA™ revenues starting in 2024. Those are continuing to 2025, but the ramps of those programs kick in post 2025. And yeah, from this perspective, I was a little bit on my comment to Tom on the booking cycle, right. We tend to know what we're building now and then for as you get further out, we have a sense of what we're launching. So, good confidence, particularly with the $5 billion of SVA™ bookings and just the $32 billion of bookings in 2022 fairly high level of confidence of what programs there are and when they'll be launching.

Mark Delaney | Goldman Sachs

That's helpful. Thanks. My second question was on Wind River. Congratulations on the Geely win that you were able to announce as you're having these conversations with your auto customers. Could you give us a bit more on the breadth of products they're looking to use? Is it sort of one or the other where maybe it's the cloud development tools, maybe it's the real-time operating system on the vehicles? Or is it pretty consistently where they're looking to use all of the Wind River solutions?

Kevin Clark | Chairman & Chief Executive Officer

It's a mixed, it's about half and half. So it's a mix of domain specific like the ADAS solution that that Matt talked about. There's several others that it's about the whole operating system, the hypervisor, as well as Wind River Studio.

David Kelley | Jefferies

Thank you. David Kelley from Jefferies. Thanks for all the color on the 2030 software revenue trajectory. I guess if we take a step back it's not an common for auto software operating margins to track in the mid-30s or even above. Is that achievable in your view on your software platform by a 2030 type timeline? And I have a follow-up.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Sure. I certainly think we'll be moving in that direction. We've talked about sort of the mid-20 plus range. So certainly tracking there obviously it's a long way out so you know comfortable will see plus 25%, how close we get to 30%, I think is in part just on what we see happening I think in part from the licensing and the subscription services. But in that zip code, yes.

David Kelley | Jefferies

Okay. Got it. Thank you. And then the SMPS product portfolio you've added capabilities really concentrated on the engineered components side dating back to I think element titan. Should we expect continued bolt-on connector, busbar, interconnect type deals? And how much of that in your view is likely concentrated in the adjacent market push going forward?

Kevin Clark | Chairman & Chief Executive Officer

I think the answer is yes. So you'll continue to see, Joe talked about the free cash flow that we will generate over the next several years. It certainly presents opportunity to do significantly more deals. You'll see more in and around the S&PS landscape, both within automotive and outside of automotive. And listen, Wind River has given us a great platform, with a 1,000 software engineers to build up our overall product portfolio from a software standpoint. So, really two strong, very strong platforms to work off of.

James Picariello | BNP Paribas

James Picariello, BNP Paribas. Slide 36, provided an important metric, I think, that directly confronts the notion of ADAS commoditization for Tier 1s. For an OEM to utilize Aptiv as the system integrated for both hardware and software, the company believes it can generate as much as 25% cost savings, right? So, my question is what's the break out between that hardware versus software? Is it about, equally split. And then just context on the reference point here for 25% lower cost versus an equivalent vision based system, like from an OEM's perspective, how do we build to this 25% savings number, I think it would be helpful.

Kevin Clark | Chairman & Chief Executive Officer

Yeah, there's more opportunity from an overall software, software reuse standpoint versus a hardware standpoint. I don't have the exact math here in front of us. But it's a mix of both, but it's more weighted to the overall software savings and driving more software reuse in a more efficient engineering development and process standpoint.

James Picariello | BNP Paribas

Okay. And then just on the 30% CAGR for high voltage electrification revenue from 2022 to 2025, I believe you're already guiding for 30% organic growth in 2023, and then you add another 15 or 20 points contribution from Intercable. So, beyond this year, I think implication will be for the revenue CAGR to kind of step down to 20% out to 25%, is that just conservatism, is it formed by your order book?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

No, the way to think that 30% we talked about this year excludes Intercable, right, because the adjusted growth rate excludes the deals until we lap a year. So, our high voltage product line, we expect to grow, to continue to grow at plus 30%. It's been sort of low 30s. Intercable has been growing in the about the mid-30% range per year and we would expect that to continue as well. As they come into that sort of – well this year obviously, they have grown that faster, but next year it will within that adjusted growth number as well.

James Picariello | BNP Paribas

Right, so then the slide that uses 2022 as the base of $1.2 billion that excluded Intercable.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Yeah.

James Picariello | BNP Paribas

So, the revenue, the implied revenue growth target for 2025 would be above just a straight 30% CAGR.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

You'd have to add in Intercable to that. Correct.

James Irwin | Moon Capital

Hi, it's James Irwin, Moon Capital. Good to see you guys. Looking at your customer exposure, following up on Adam's earlier question, in the last two years in China, just massive share erosion of traditional legacy, foreign JVs, Volkswagen, GM. So, five years from now, if a BYD, a Geely have tremendous global market share growth, does that change your forecasts that dramatically? Because I get the impression you're leveraging the majority of your technology and it's not that customer-specific. So, whether it's a manufacturing footprint, your ability to switch to whether your customers are growing, you can adapt to that. I just want to make sure that assumption is not too optimistic?

Kevin Clark | Chairman & Chief Executive Officer

No, it's not. I mean, today, roughly from a revenue standpoint, we're a little less than 50/50 in the China market with the locals versus the multinationals. When you look at our bookings and trend in our bookings, it's more towards the locals. So, that natural weighting, to the extent they grow outside of the European market, I'd have to think about who they're taking market share from.

But, just given our overall position, I think we'd be well-positioned to certainly benefit or not be hurt by that as it relates to resources in China to develop solutions with the China market. As you know, Jim, we've talked about this for a long time. We've been in China for 25-plus years and developing whether it's solutions for our S&PS business or AS&UX business, developing solutions in China, for China and that's a market when you look at, especially on the AS&UX side, is actually evolving faster than what you see in Europe and in North America.

James Irwin | Moon Capital

And you touched on, the second question. You touched on the regionalization, I think moving from moving towards 95% just a few years and I appreciate all the color on Wind River. That single pane approach that you mentioned, it's kind of simplifying it for a guy like me. When you have China, it seems like every other day a balloon's are being shot out of the sky now, a lot of political tension. From a regulatory standpoint, does that single pane approach that Wind River brings to the table, can that be regionalized as well?

Kevin Clark | Chairman & Chief Executive Officer

Yeah. Our path on that is that single pane in China is for China. And that single pane outside of China is for effectively Western, whether it's European or US.

Chris McNally | Evercore ISI

Thanks team, Chris McNally from Evercore. Joe, a quick numbers question on the third leg of the AS&UX tool that you're calling Smart Compute and Software. When I put the numbers together from a couple of the slides, $500 million of revenue, now it seems like $1.1 billion on a 30% growth 2025. If you sort of look forward at year 2030, it seems like that's like a $7 billion sort of plus number just looking at the pie chart. We know Wind River, is the rest of that mostly zonal controllers?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

SVA™ will start, yeah, I think I said $6 billion in the prepared remarks is our current estimate, with obviously within the estimate $40. Yeah, the remainder will be SVA™ revenue at this point. So, that product line, and we will report out, andwe got a lot of questions about, sort of providing that visibility go forward. So, while we figure that product line was the best way to do it, so that will be the Wind River number as well as effectively SVA™ compute.

Chris McNally | Evercore ISI

And then the software would that be in, in S&PS, because I know at one point you were talking about splitting up some of the SVA™ revenue into two parts. But so just want to be clear on that.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

Software around Smart Vehicle Compute and that's why we changed the name a little bit. Just to be clear, that'll be the, that'll be the compute side of SVA™. So, that will be the compute, the software as well as on SVA™ as well as Wind River. SVA™ revenue is around the nervous system of the business, will be part of the S&PS segment.

Chris McNally | Evercore ISI

SPS, perfect. And then just last on the visibility of SVA™, if you just remind everyone the updated OEM number, how many OEMs you are in sort of current discussions with or booked business just so we can sort of.

Kevin Clark | Chairman & Chief Executive Officer

Well current discussions with, listen, we talked about launching SVA™ and the concept of SVA™ since 2017. In terms of discussions, I'm not sure it's dozens, but it's a significant number of OEMs where we have a tremendous amount of visibility to what they're doing from an overall roadmap as it relates to current wins and near-term discussions.

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

So to Kevin's point discussions, a lot of OEs. Formal advanced development agreements, 11, and Wins that are in that $5 billion worth of bookings, seven OEs.

Chris McNally | Evercore ISI

Seven OEs. If I could just squeeze one last more, you have talked about the potential for SVA™ content per vehicle in that 2030 $6 billion number, what an average sort of CPV would be your booked business as opposed to what like the potential could be?

Joe Massaro | Chief Financial Officer & Senior Vice President, Business Operations

For 2030, it would be part of Kevin's chart that shows the $2,300 average for that period.

Kevin Clark | Chairman & Chief Executive Officer

It would be a subset.

Audience Question

It's great to see details on sustainability with a separate session today. You talked about a target of 25% carbon reduction on Scope 1 and Scope 2. I'm wondering whether you think about integrating Scope 3 in your target. And in general, when we look at sustainability strategy, what are the main challenges for Aptiv to have even more ambitious strategies over long term? Thank you.

Kevin Clark | Chairman & Chief Executive Officer

Just want to make sure we understand your question. You're looking for what is our strategy as it relates to Scope 3 and how do we become more ambitious? As you know, Scope 3 requires that we're closely integrated with our customers, obviously with ourselves and with our broader supply chain, their OEMs, especially in Europe that have much more aggressive objectives and adaptions to be net zero carbon by 2030. So, those are the players, the extent we are aligned from a customer standpoint, it's easier for us to align with our overall supply base standpoint. Internally we are positioned to flex in whatever directions required. I don't know Kate, if you want to add anything to that.

Kate Ramundo | Senior Vice President, Chief Legal Officer, Chief Compliance Officer & Secretary

No, I agree with that, Kevin.

Kevin Clark | Chairman & Chief Executive Officer

Any more questions? Any virtual questions.

Jessica Kourakos | Vice President, Investor Relations & ESG

We had one and it was already addressed. So, I think we can conclude with whatever remarks you would like to make Kevin.

Kevin Clark | Chairman & Chief Executive Officer

Okay. Well, listen, one, we appreciate everybody joining us today. We recognize that it is Valentine's Day, thanks for spending a part of your day with us. We appreciate you as investors and shareholders and know that we will follow up this meeting with other discussions with you to get you even more informed about where our industry is headed and where Aptiv sits and benefits from those trends. So, thank you again. Take care.